

Urban Economic Structure and Disaster Resilience

Labor Markets, Road Networks, and Trade Linkages in Disaster Resilience

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2026-08-14

Abstract

Natural disasters usually have severe economic consequences in urban areas because the concentration of economic activity amplifies agglomeration’s vulnerability. However, cities differ in their economic structures, including industrial composition, road networks, and trade linkages. These different configurations of agglomeration generate distinct patterns of vulnerability to natural disasters, revealing substantial heterogeneity masked by the average 3 percent decline in aggregate economic activity at the metropolitan statistical area (MSA) level. Having fewer industries more than doubles losses, whereas more centralized road networks and stronger internal goods sourcing mitigate losses by around half. I also uncover heterogeneity across industry groups, with a greater manufacturing share mitigating losses and larger accommodation, trade, and transportation (ATT) and health and education (HE) shares amplifying them.

Cities concentrate economic activity within a confined space. In the United States, metropolitan statistical areas (MSAs) account for the vast majority of employer establishments and population¹, thereby serving as the primary locus of production, employment, and innovation. This greater geographic concentration forms the foundation of agglomeration economies, yielding greater productivity. The same agglomeration of economic activities, nonetheless, also provides the foundation for greater damages from natural disasters. Damages are generally greater in urban regions, with the overall impacts of natural disasters further amplified by their increasing frequency and intensity². Thus, agglomeration is a source of productivity gains during good times, but also a source of

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¹In 2023, MSAs contained approximately 7.9 million of the nation’s 8.4 million employer establishments, accounting for nearly 95 percent of all employer establishments in the United States. Likewise, approximately 86 percent of the U.S. population, 294 million people, resided in MSAs as of July 1, 2024.

²For instance, there has been a roughly fourfold increase in billion-dollar disasters since the 1980s. NOAA National Centers for Environmental Information, U.S. Billion-Dollar Weather and Climate Disasters (2024), <https://www.ncei.noaa.gov/access/billions/>, DOI: 10.25921/stkw-7w73.

vulnerability during bad times, a dimension that has received far less attention.

A further and much less understood aspect of the costs of agglomeration in the context of natural disasters is that the realized disruptions caused by natural disasters depend on a city’s underlying configuration of economic activities. A city’s employment may be concentrated in a narrow set of industries, its road network may depend on a limited number of critical road segments, and its goods sourcing may rely primarily on local suppliers. Together, these three dimensions characterize how workers, transportation, and goods are organized within cities. From this perspective, each city’s specific configuration of industrial composition, road networks, and trade linkages may play an important role in shaping the magnitude and persistence of economic disruptions following natural disasters, beyond what can be attributed to disaster intensity alone. Differences in these urban structural configurations therefore constitute an important source of heterogeneity in urban resilience. However, despite extensive attention to the productivity benefits of agglomeration³, relatively little is known about how these multiple dimensions of urban economic structure amplify or mitigate economic disruptions following natural disasters.

This paper examines whether differences in industrial composition, road networks, and trade linkages strengthen or weaken the economic consequences of geographic concentration following natural disasters. In doing so, it moves beyond evaluating urban economic organization solely through the lens of productivity under ordinary economic conditions to examine how alternative forms of urban economic structure shape heterogeneity in resilience to natural disasters. Ultimately, it asks under what structural conditions agglomeration remains an asset rather than becoming a source of vulnerability, thereby shaping how geographic concentration translates into economic losses following natural disasters. Identifying the heterogeneous ways in which different dimensions of urban economic structure shape disaster impacts requires a carefully designed empirical strategy, given

³Regarding the labor market concentration, Glaeser et al. (2004) and Behrens et al. (2014) emphasize the fundamental role of skilled human capital in city development. Wage premia arising from firm competition in cities attract additional workers, further expanding urban areas and reinforcing agglomeration economies. Duranton et al. (2001) illustrate the dynamics of city-level specialization, showing that innovation tends to emerge in diversified cities, while mass production often relocates to more specialized cities with lower production costs. In this paper, city diversity is measured as the inverse of the Herfindahl index based on sectoral employment shares within each city.

Regarding transportation infrastructure, Duranton et al. (2012) find that a greater initial stock of highways leads to higher long-run employment and faster economic growth at the city level. Similarly, Baum-Snow (2007) shows that the expansion of the interstate highway system contributed substantially to urban decentralization in the United States between 1950 and 1990.

From the perspective of economic geography and trade, Marin et al. (2021) document that export activity is disproportionately concentrated in larger cities, implying greater potential gains from trade at the urban level and reciprocal benefits for city development.

Furthermore, three agglomeration forces are also closely interconnected. For example, Marin et al. (2021) show that active trade shifts labor supply toward larger cities, while Duranton et al. (2014) find that interstate highways increase the overall volume of exports across U.S. cities, though they have little effect on total export value.

the inherently unpredictable and transient nature of natural disasters. To address this challenge, I employ the local projection difference-in-differences (LP-DiD) framework developed by Dube et al. (2023) and Jordà et al. (2025) using a monthly panel of U.S. MSAs spanning 2018 to 2024. The LP-DiD framework accommodates intermittent exposure to natural disasters while constructing appropriate control cities that remain unexposed within each event window. The LP-DiD framework is particularly beneficial for accounting for the temporary and recurring nature of natural disasters, which can affect the same city multiple times over the period.

I use monthly commercial and industrial electricity consumption at the MSA level as a proxy for local economic activity, following Cicala (2023). This measure captures broad economic activity within a city at a high frequency, making it particularly well-suited to measuring its immediate response to disruptions caused by natural disasters. Nonetheless, because electricity consumption may raise concerns about its representativeness as a proxy for overall economic activity, I also examine an alternative labor market outcome using monthly employment changes. Although employment captures a narrower aspect of economic activity than electricity consumption, it provides a direct measure of labor market conditions and serves as a robustness check for the main findings. To obtain an objective measure of disaster intensity independent of realized economic damages, I use average monthly wind speeds, with treatment defined as wind speeds corresponding to Category 2 hurricanes or higher.

This paper yields three main findings. First, natural disasters with Category 2 or higher intensity lead to substantial short-run economic declines. For a city in which no one structural feature is particularly pronounced, a high-wind disaster reduces monthly aggregate economic activity by approximately 3 percent relative to pre-disaster levels. Consistent with the decline in electricity consumption, average employment over the subsequent three months decreases by roughly 600 hires, indicating substantial disruptions to local production activity.

Nonetheless, the average effects mask substantial heterogeneity across cities arising from differences in urban organization. Across the three dimensions of labor market composition, road network structure, and trade linkages, I show that cities with more concentrated labor markets experience larger economic disruptions, whereas those with more centralized road networks and stronger intra-city goods-sourcing capacity exhibit greater resilience to natural disasters. More specifically, a one-standard-deviation increase in industrial concentration deepens the decline in economic activity from 3 percent to approximately 7.3 percent. Put differently, when employment is concentrated in a narrow set of industries rather than distributed across a more diversified labor market, the

short-run economic disruptions from natural disasters become roughly 2.5 times larger than the baseline effect. In contrast, a one-standard-deviation increase in road network centralization reduces the post-disaster decline in economic activity by approximately 60 percent. Cities in which a smaller number of critical road segments play a central role in connecting locations therefore experience substantially smaller economic disruptions following natural disasters. Similarly, a one-standard-deviation increase in intra-city goods-sourcing capacity reduces the post-disaster decline in economic activity by approximately 44 percent. When a city relies more heavily on goods sourced from within the city rather than from external suppliers, the resulting short-run economic disruptions are substantially smaller. Taken together, these findings underscore the heterogeneous roles of urban economic structures in shaping disaster resilience, with different forms of concentration either amplifying or mitigating economic damages.

Finally, I find that the vulnerability associated with industrial composition further depends on the specific industries in which employment is concentrated. I distinguish six industry groups: Retail, Utilities, and Construction (RUC); Business Services (BS); Health and Education (HE); Manufacturing; Accommodation, Wholesale Trade, and Transportation (ATT); and Others. From an industry-group-specific perspective, a city’s overall labor market concentration by industrial composition can be decomposed into two distinct components: a group’s share of total city employment (size) and the concentration of employment across detailed industries within the group (sectoral specialization)⁴. I first show that Manufacturing, ATT, and HE contribute distinctly to disaster vulnerability. Specifically, a larger employment share in Manufacturing mitigates the economic impacts of natural disasters, whereas a larger employment share in ATT and HE exacerbates them. Furthermore, I find that only HE exhibits an additional effect of sectoral specialization, implying that its exacerbating effect arises not only from a larger employment share but also from employment being concentrated in a small number of detailed industries within the group. These findings suggest that the resilience consequences of industrial concentration differ across industry groups. Furthermore, the impacts of industrial concentration depend not only on its scale and intensity but also on the industry group in which that concentration occurs, with important implications for evaluating urban specialization and ongoing structural transformation under increasing disaster risk. Taken together, these findings suggest that the benefits of agglomeration cannot be evaluated solely through productivity, as organizational structures that perform well under normal conditions

⁴ Accordingly, a city’s overall labor market concentration may be high because a particular industry group accounts for a large share of total employment, because employment within that group is concentrated in a small number of detailed industries, or because of both.

may be considerably more vulnerable to natural disasters.

The main contribution of this paper is to identify the distinct roles that different dimensions of urban economic structure play in shaping the economic impacts of natural disasters. From the perspective of industrial composition, natural disasters provide a setting in which the long-standing debate between specialization and diversification appears to favor diversification, as less diversified local economies experience more severe economic damages (Heblich et al. (2025); Moretti et al. (2024); Combes et al. (2012); Duranton et al. (2001); Glaeser et al. (1992); Jacobs (1969); Marshall (1890)). The mitigating effects of more centralized road networks may reflect the faster restoration of critical transportation infrastructure (Miyauchi et al. (2025); Bergantino et al. (2024); Allen et al. (2022); Tyndall (2021); Fajgelbaum et al. (2020); Duranton et al. (2012)), while stronger intra-city goods-sourcing capacity may operate through thicker buyer–supplier linkages that help sustain economic activity in the aftermath of a disaster (Miyauchi (2024); Marin et al. (2021); Duranton et al. (2014)).

Another contribution of this paper is to examine heterogeneity in disaster resilience across industry groups by highlighting the role of industry-group-specific resilience to natural disasters, an aspect that has received little attention in the existing literature. While the importance of cognitive and non-routine intensive industries including of HE and BS in promoting urban growth is well established (Monte et al. (2023); Eckert et al. (2022); Hsieh et al. (2023); Jarosch et al. (2021); Rossi-Hansberg et al. (2021); Fajgelbaum et al. (2020); Rossi-Hansberg et al. (2019)), far less is known about whether resilience to natural disasters differs systematically across industry groups. This paper fills that gap by showing that concentration in Manufacturing, ATT, and HE contributes distinctly to disaster-related economic losses, with each industry group exhibiting effects that differ not only in magnitude but also in direction.

The remainder of the paper is organized as follows. The next section presents the conceptual framework for understanding the multidimensional aspects of agglomeration. Section II describes the construction of the monthly economic activity proxy, the measure of natural disaster intensity, and the three measures of urban economic structure. Section III presents the empirical context and research design, with particular attention to the LP-DiD framework. Section IV presents the main empirical results. Section V examines heterogeneity in disaster resilience across detailed industry groups. Section VI provides supporting evidence on the mitigating role of centralized road networks in natural disasters. Section VII presents results using alternative outcome measures, and Section VIII reports a range of robustness checks. Section IX concludes.

I Conceptual Framework

Agglomeration is a fundamental feature of economic activity. By concentrating firms, workers, and production within geographic space, agglomeration generates substantial productivity gains. Yet the degree of spatial concentration alone does not fully capture the organization of economic activity. Cities differ in the underlying economic structures through which agglomeration is realized, so cities with similar levels of agglomeration may exhibit fundamentally different patterns of economic organization. A city’s overall degree of spatial concentration is well represented by population density. Beyond this overall concentration, industrial composition, road networks, and trade linkages capture distinct dimensions of urban economic structure. Cities differ not only in overall agglomeration intensity but also in the underlying dimensions of urban economic structure. Figure 1 displays the standardized values of labor market concentration, transportation network centrality, and within-city goods sourcing capacity across all 365 MSAs in the sample, with six illustrative cases highlighted.

A comparison of the Midland, TX, and Las Vegas, NV, MSAs illustrates their differences in overall agglomeration intensity, as measured by population density. Nevertheless, the two MSAs share similar labor market structures. Specifically, Midland is among the least densely populated metropolitan areas in the United States, whereas Las Vegas is among the most densely populated. Despite this stark contrast in overall agglomeration, both MSAs exhibit similarly high levels of industrial concentration. In each city, employment is concentrated in a relatively small number of industries, although the dominant industries differ substantially. Midland’s economy is heavily concentrated in oil and gas activities, whereas Las Vegas specializes in entertainment, hospitality, and tourism. Those two MSAs, however, also diverge along the road network and trade dimensions of agglomeration. Las Vegas exhibits a substantially higher degree of within-city goods sourcing, indicating a greater capacity for local production and exchange, while Midland shows much lower within-city sourcing. The pattern is reversed for road networks. Midland has a more centralized transportation network, with travel concentrated along a limited number of major corridors. By contrast, Las Vegas has a more dispersed road network that provides multiple alternative routes, resulting in lower transportation network centrality. This comparison illustrates that differences in overall spatial concentration alone do not fully characterize agglomeration. Even cities with markedly different levels of agglomeration may share important structural characteristics while differing substantially in others. This comparison illustrates that differences in overall spatial con-

centration do not preclude similarities in some dimensions of urban structure while revealing marked differences in others.

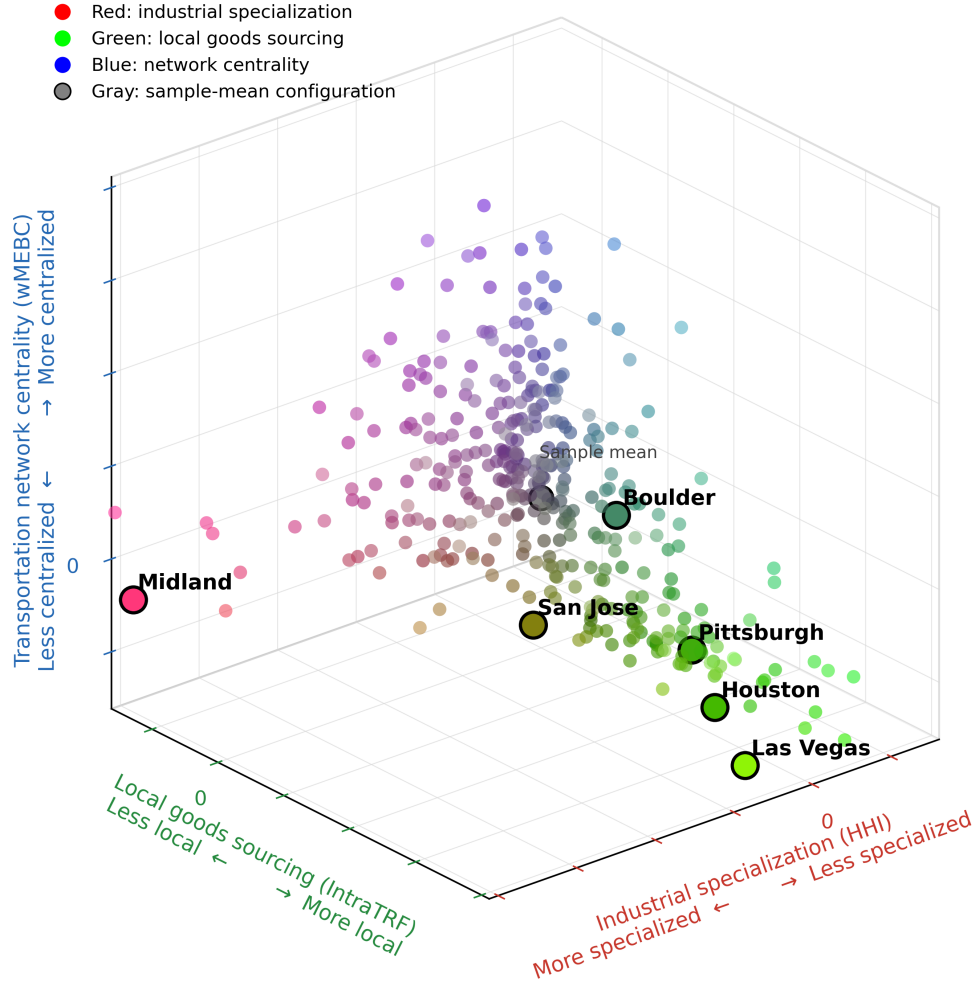
The converse is equally true. Similar levels of agglomeration are compatible with both similarities and differences in the underlying organization of economic activity. Consider the San Jose, California, and Houston, Texas, MSAs. The two cities exhibit similarly high levels of overall agglomeration, relatively dispersed road networks, and strong within-city goods sourcing capacity. Despite these similarities, they differ sharply in industrial composition. San Jose exhibits a high degree of industrial concentration, with employment clustered in high-technology industries and producer services, including finance and professional services. Houston, by contrast, has a much more diversified employment base, with no small group of industries accounting for a dominant share of local employment. A parallel comparison between the Pittsburgh, PA, and Boulder, CO, MSAs isolates differences in transportation network organization. The two metropolitan areas exhibit similar levels of overall agglomeration, industrial composition, and within-city goods sourcing. Despite these similarities, they differ sharply in their road networks. Pittsburgh’s road system provides numerous alternative routes between locations, whereas Boulder’s network is constrained by the surrounding topography, concentrating travel along a smaller number of critical corridors. Consequently, Boulder exhibits substantially higher transportation network centrality.

Taken together, these comparisons illustrate that urban heterogeneity extends well beyond aggregate density. Cities may resemble one another along some dimensions of urban organization while differing substantially along others. Aggregate concentration, labor market organization, transportation network structure, and local goods-market connectivity therefore capture distinct dimensions of urban economic structure, each of which may vary independently across cities. These differences in how economic activity is organized imply that cities need not respond uniformly to external shocks, making each city’s organizational configuration a potential source of heterogeneous resilience to natural disasters. Indeed, cities exposed to natural disasters of comparable intensity often experience markedly different short-run economic disruptions. These differences suggest that the consequences of natural disasters depend not only on the characteristics of the shock itself but also on the underlying organization of the affected local economy⁵.

⁵Illustrative examples include the widespread adoption of remote work during recent nor’easters, which allowed many office-based activities to continue despite hazardous travel conditions (*The Boston Globe*, “Nor’easter snowstorm forces many workers remote while essential employees continue commuting,” February 24, 2026.); repeated closures of Interstate 80 during winter storms that disrupted freight movement because of limited alternative routes *Associated Press coverage of widespread grocery shortages and panic buying ahead of major winter storms*; and panic buying ahead of major winter storms that temporarily strained local retail and distribution systems (*Reuters*, “Panic buying as people rush to stock up ahead of winter storm,” January 2026).

Figure 1: Cities Differ in How Economic Activity Is Organized

Dimensions of Urban Economic Organization



Notes: Each point represents a U.S. MSA. The three axes summarize distinct dimensions of urban organization: industrial concentration (HHI), road-network structure (weighted edge betweenness centrality; $wMEBC$), and intra-city trade ($IntraTRF$). Because all three measures are standardized, zero on each axis denotes the sample mean of that dimension, while the gray point at the origin represents the joint sample-mean configuration. Point colors provide an additional representation of each MSA's configuration across the three dimensions: red, green, and blue intensities correspond to HHI , $IntraTRF$, and $wMEBC$, respectively, with darker (brighter) intensities indicating values further below (above) the sample mean. The resulting color mixture therefore reflects each MSA's multidimensional urban configuration. Selected MSAs illustrate that cities with similar characteristics along one dimension may differ substantially along others, motivating the multidimensional characterization of urban structure used throughout the paper.

II Data

II.I Electricity Consumption Loads

Electricity consumption serves as a high-frequency proxy for local economic activity as it is an essential intermediate input into virtually all production and commercial activities⁶. Changes in economic activities are therefore reflected almost immediately in electricity demand, allowing the measure to capture short-run fluctuations following natural disasters. More specifically, I focus on industrial and commercial electricity consumption, excluding residential and transportation usage, to better capture general economic activity.

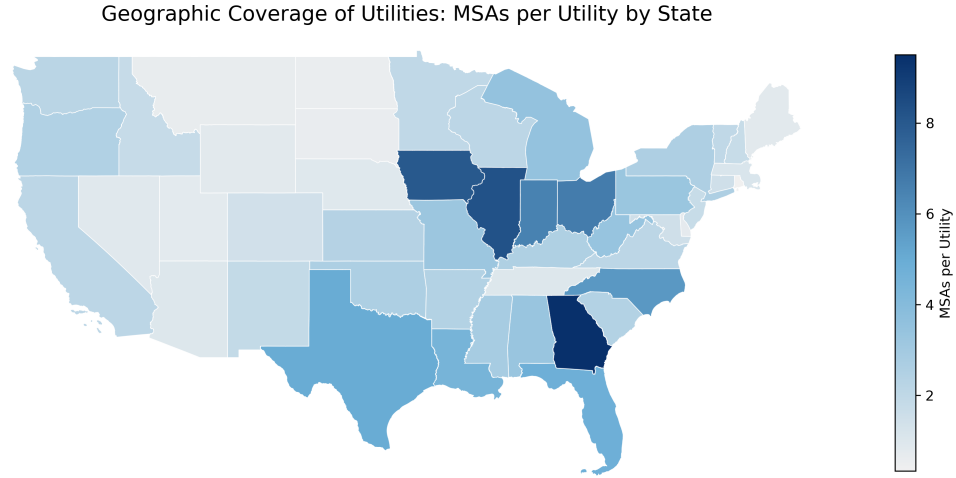
More conventional measures of aggregate economic activity, such as gross domestic product (GDP), are available only at the state level at a quarterly frequency and are therefore too spatially aggregated to capture agglomeration effects at the metropolitan level. Employment-based measures are available monthly at the MSA level, but primarily capture adjustments along the labor-market margin and may respond more gradually to short-run disruptions in underlying economic activity. Electricity consumption therefore provides a useful combination of temporal frequency and spatial granularity for examining the immediate economic effects of natural disasters. I discuss this distinction in greater detail and examine employment-based outcomes as an alternative measure in Section II.I.2.

I use the U.S. Energy Information Administration’s (EIA) Form 861M, the Monthly Electric Power Industry Report, and Form 861, the Annual Electric Power Industry Report. Electricity sales data are available at the utility-month level while a utility’s service territory frequently spans multiple counties and may extend across multiple MSAs and state boundaries. Consequently, utility service territories do not generally align with MSA boundaries, requiring the construction of MSA-level measures from utility-level electricity sales data. Figure 2 presents the geographic scope of utilities varies substantially across states. On average, a single utility within one state serves approximately 2.7 MSAs, with a median of 2.1 and a range from 0.3 to 9.5 MSAs.

To construct MSA-level electricity consumption measures, I combine utility-level sales data with the Service Territory Information schedule from EIA Form 861, which identifies the counties served by each utility. I then allocate utility-level electricity sales across counties using county population weights and aggregate the resulting county-level estimates to the MSA level. As a robustness

⁶Cicala (2023) demonstrates that electricity consumption across industrial, commercial, transportation, and residential sectors closely tracks economic activity, particularly during the COVID-19 pandemic.

Figure 2: Geographic Coverage of Utilities: MSAs per Utility by State



Note: Darker shades indicate that a single utility serves a larger number of MSAs within a state.

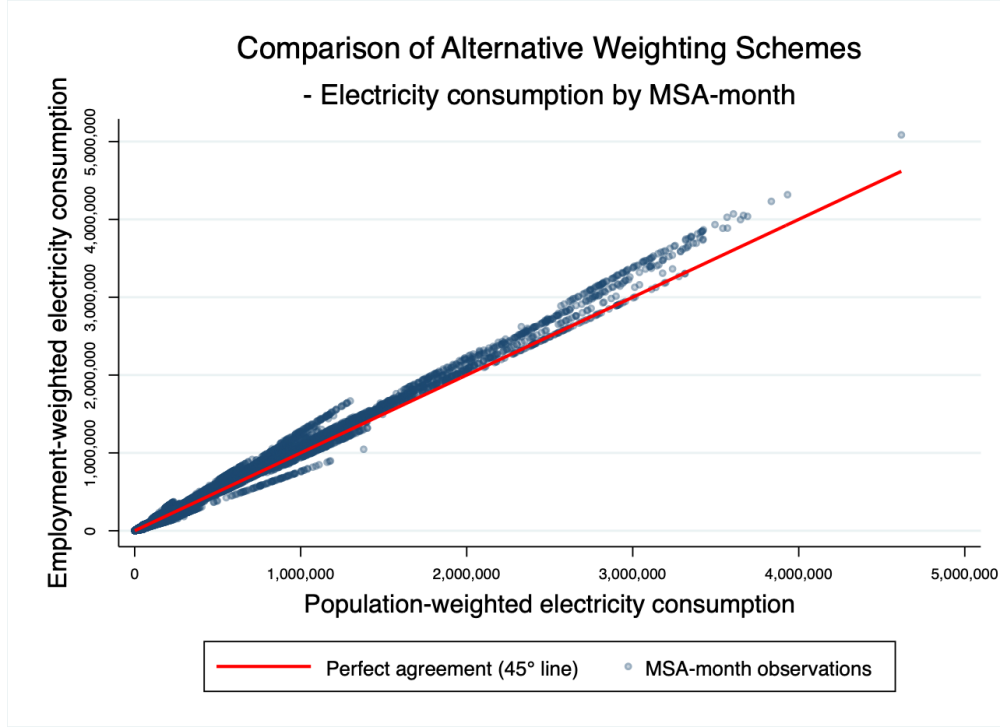
check, I also construct MSA-level electricity consumption using county employment shares from the County Business Patterns (CBP) as an alternative weighting scheme. Figure 3 compares electricity consumption measures constructed using population and employment weights. The two measures exhibit a close one-to-one relationship, indicating that the electricity consumption measure is largely robust to the choice of weighting scheme.

II.I.1 Additional Controls for Electricity Consumption

When using electricity consumption as a proxy for MSA-month-level aggregate economic activity to estimate the economic effects of natural disasters, it is important to account for two external factors that directly affect electricity consumption, namely physical damage to the electricity system and climatological conditions. For instance, disaster-induced damage to the electricity transmission and distribution network may reduce electricity usage independently of changes in underlying economic activity, potentially confounding the estimated effects of natural disasters. Likewise, unusually high or low temperatures can increase electricity demand for cooling or heating independently of the economic impacts of natural disasters, potentially introducing additional confounding variation into electricity consumption.

To capture the physical damage aspect of natural disasters, I measure outage intensity using the average daily outage observation count at the MSA-month level. I use electricity outage data from the Environment for the Analysis of Geo-Located Energy Information (EAGLE-I) program

Figure 3: Comparison of Electricity Consumption Measures under Alternative Weighting Schemes



Note: Each point represents an MSA-month observation. The red 45° line indicates perfect agreement between the two weighting schemes. The close alignment of observations with the 45° line suggests that the electricity consumption measure is largely insensitive to whether population or employment weights are used.

developed by Oak Ridge National Laboratory (ORNL) for the U.S. Department of Energy (DOE) to account for this channel. The dataset provides county-level outage information from 2014 to 2025, recording the number of customers without power during outage events. Specifically, I count the number of recorded outage observations within each county-month and divide this count by the number of calendar days in the month to obtain the county-level average daily outage observation count. I then aggregate this measure to the MSA-month level by taking the mean across counties within each MSA. To account for climatological effects on electricity consumption, I also include heating and cooling degree days obtained from the National Oceanic and Atmospheric Administration (NOAA). Finally, I include the MSA-month average share of buildings constructed before 1980 to account for differences in the building stock from the U.S. Census Bureau’s American Community Survey (ACS). Older buildings tend to be more vulnerable to natural disasters, potentially amplifying the effects of physical damage on electricity consumption. At the same time, building vintage may influence the responsiveness of electricity demand to weather conditions, as newer buildings are more likely to have features such as air conditioning and better insulation.

II.I.2 Alternative Outcomes

As an alternative measure of economic activity from the labor market, I use MSA-month-level three-month average employment growth from the Current Employment Statistics–State and Area Employment (CES-SAE) program⁷, although labor market outcomes capture only labor adjustment following natural disasters and are therefore narrower than electricity consumption as a measure of aggregate economic activity. A key consideration is that changes in employment, rather than employment levels, are more informative for capturing the short-run economic disruptions caused by natural disasters. This emphasis is consistent with Gentili (2025), who measure the immediate economic impacts of the May 2023 floods in Italy using changes in high-frequency indicators of economic activity of new daily hires.

II.II Natural Disaster Intensity

I focus on natural disasters associated with high winds and measure disaster intensity using observed wind speeds. Unlike damage-based monetary measures, observed wind speeds are not mechanically influenced by local economic activity or the value of exposed assets, making them an objective measure of the underlying hazard intensity. Moreover, strong winds can exacerbate damages across a wide range of natural disasters, including hurricanes, floods, and wildfires, making wind speed a useful and objective indicator of disaster severity. The primary data sources for disaster events are the Storm Events Database maintained by NOAA’s National Centers for Environmental Information (NCEI) and the Atlantic Hurricane Database (HURDAT2) provided by NOAA’s National Hurricane Center (NHC). In the Storm Events Database, recorded wind speeds are available for event types including High Wind, Marine High Wind, Marine Strong Wind, Marine Thunderstorm Wind, Strong Wind, and Thunderstorm Wind, which are classified separately from tropical storms, hurricanes, tornadoes, floods, and hail. For Atlantic hurricanes, the HURDAT2 database provides sustained maximum wind speeds, which I use as the measure of disaster intensity. I exclude tornado events from the analysis because direct wind speed measurements are not available for tornadoes⁸.

Using MSA-level monthly average wind speed, I construct an indicator for months in which average wind speed exceeds 96 mph, corresponding to the lower bound of a Category 2 hurricane

⁷The QCEW provides near-universal coverage of employment and wages based on unemployment insurance records, making it a comprehensive measure of local labor market conditions. In contrast, the CES-SAE program offers more timely indicators of labor market dynamics, particularly for nonfarm employment.

⁸Tornado intensity is classified using the Enhanced Fujita (EF) scale, which infers wind speed ranges based on observed post-event damage rather than direct measurements. Because these values are inferred rather than observed, I exclude tornado exposures from the analysis.

and thus capturing events from Category 2 through Category 5. Monthly average wind intensity at the MSA level is preferred to monthly maximum wind intensity because it is a more conservative measure and is less susceptible to the timing of natural disasters within a month. In particular, average wind intensity provides a more stable measure of monthly exposure regardless of when a sufficiently strong wind event occurs, whereas maximum wind intensity depends entirely on the single strongest observation during the month. Table 1 summarizes the Saffir–Simpson hurricane wind scale and the corresponding approximate damage classifications. Table 2 reports the monthly frequency of exposure to Category 1–5 hurricane-force winds between 2018 and 2024. The frequency of exposure declines sharply for Category 2 and higher winds, although hurricane exposure remains broadly distributed throughout the year.

Table 1: Saffir–Simpson Hurricane Wind Scale Categories

Category	Wind Speed (mph)	Wind Speed (knots)	Damage Potential
Category 1	74–95	64–82	Minimal to moderate damage
Category 2	96–110	83–95	Moderate to extensive damage
Category 3	111–129	96–112	Devastating damage
Category 4	130–156	113–136	Catastrophic damage
Category 5	≥ 157	≥ 137	Catastrophic to total devastation

Table 2: Monthly Frequency of High-Wind Storms by Category 1–5 (2018–2024)

Month	Cat 1	Cat 2	Cat 3	Cat 4	Cat 5
Jan	54	6	1		
Feb	36	2	2		
Mar	44	2	1		
Apr	43	2			
May	25	1			
Jun	29	1			
Jul	27	1		1	
Aug	30	5	1	1	
Sep	23	3		1	
Oct	26	2	1	1	1
Nov	29	1			
Dec	46	2	1		
Total	412	28	7	4	1

II.III Urban Economic Structures

I construct three key dimensions of urban agglomeration from pre-period measures calculated using 2017 data. First, to capture the concentration of employment across industries within a city, I calculate the Herfindahl–Hirschman Index (HHI) of industrial composition. Second, to capture overall road network centrality after accounting for the observed commuting patterns, I compute the weighted Mean Edge Betweenness Centrality ($wMEBC$). Finally, to measure the extent to which a city relies on internal sourcing of goods, I construct the Intra-city Trade Flow Ratio ($IntraTFR$).

Labor Market Job Concentration

First, regarding industrial composition, the key question is how concentrated employment is across industries within a city. The Herfindahl–Hirschman Index (HHI) of the labor market measures the degree of employment concentration across industries within a city. MSA-month-industry-wise job counts are from the Quarterly Census of Employment and Wages (QCEW), which provides industry-specific employment data classified under the North American Industry Classification System (NAICS). Offered job counts are summarized at the NAICS code 4-digit, industry group level. As a result, the HHI of the labor market is defined as follows:

$$\begin{aligned} HHI_{c,m,y} &= \sum_{i=1}^N s_{i,c,m,y}^2 \\ &= \sum_{i=1}^N \left(\frac{J_{i,c,m,y}}{J_{c,m,y}} \right)^2 \end{aligned} \quad (1)$$

where N represents the total number of industries in city c , $s_{i,c}^2$ represents the employment share of industry i in city c , month m and year y . To be more specific, $J_{i,c,m,y}$ represents the number of jobs in industry i and $J_{c,m,y}$ represents the total number of jobs across all industries. The 2017 average HHI , which ranges from 0 to 1, measures the degree of industry concentration, with higher values indicating greater industrial concentration and lower employment diversity. For example, the Midland MSA in Texas has the second-highest average HHI among all MSAs, reflecting a highly concentrated labor market with limited industry diversification. This pattern is driven by the concentration of employment in the oil and gas sector and related industries, which together account for more than 50 percent of the local workforce⁹.

⁹Midland MSA is home to major oil companies, including EOG Resources, Concho Resources, Pioneer Natural Resources, Chevron, Shell, ExxonMobil, and Occidental Petroleum.

Road Network Concentration

For the road network, I model road segments as edges and intersections as nodes in the entire network, and evaluate the importance of each road segment based on its role in connecting locations throughout the network (Marshall (2016) Marshall (2016)). A representative measure widely used in the transportation (Sakakibara et al. (2004)) and social network (Qi et al. (2017)) literature is Edge Betweenness Centrality (EBC), which captures the degree of road network centralization. More specifically, edge centrality is determined by the extent to which a road segment lies on the shortest paths between pairs of locations. Road segments with higher EBC are traversed by a larger share of shortest paths, indicating that fewer alternative shortest routes exist and that these segments are more critical for efficient travel across the network. For example, when a road segment constitutes the only shortest path between two locations within a city, its disruption forces travelers onto longer alternative routes, thereby reducing the efficiency of travel throughout the network.

In addition to the physical structure of the road network, recent literature highlights the role of commuting patterns in shaping localized economic outcomes (Monte et al. (2018)¹⁰). Workers in on-site occupations that are difficult to perform remotely must rely on the road network to access their workplaces, whereas workers in telecommutable occupations, such as financial services, face substantially weaker commuting constraints. Recognizing that commuting burdens vary with occupational composition, I extend the $MEBC$ measure to incorporate the commuting patterns of workers who are physically required to work on site. Under this framework, weighted road network centrality reflects the extent to which an MSA’s transportation network depends on a small number of critical road segments, especially for commuting.

Census block level commuting patterns by occupation are drawn from the LEHD Origin Destination Employment Statistics (LODES), which provide home-workplace block pairs across three industry classifications including (1) goods-producing, (2) trade, transportation, and utilities, and (3) all other services. Specifically, I treat each MSA as the destination of commuters, with each commuter assigned to a workplace located in a census block. For each census block, I observe the occupational composition of arriving commuters and use this information to weight road network centrality by the share of workers who are physically constrained to work on site. Among various occupations, I classify jobs in goods-producing industries, trade, transportation, and utilities as

¹⁰Monte et al. (2018) show that variation in commuting patterns serves as an adjustment margin through which the localized effects of shocks are redistributed across space. In particular, heterogeneous commuting patterns across counties play a key role in shaping local employment responses to labor demand shocks.

more physically constrained to the worksite¹¹. This classification enables the measure to capture heterogeneity in commuting intensity and physical presence requirements across occupations.

Based on this concept, I construct a commuting pattern adjusted road-network centrality at the block level. First, to compute topological indices, I use road networks obtained from OpenStreetMap (OSM) using the OSMnx module in Python (Boeing, 2017a Boeing (2017a)). Let b index workplace census blocks, and let $\mathcal{B}_c$ denote the set of workplace blocks located in MSA c . I begin by assigning to each block b a measure of road-network centrality derived from the metropolitan road network N_c .

$EBC_b(N_c)$ denote the block-assigned road-network centrality, defined as the average edge-betweenness centrality of road segments in the metropolitan road network N_c that are spatially associated with workplace block b :

$$EBC_b(N_c) = \frac{1}{|E_b(N_c)|} \sum_{e \in E_b(N_c)} EBC(e; N_c) \quad (2)$$

where $E_b(N_c) \subseteq E(N_c)$ denotes the set of road segments in N_c assigned to block b through the spatial overlay procedure.

Using this block-level assignment, I construct a commute-adjusted measure of road-network exposure at the block level:

$$wEBC_b = EBC_b(N_c) \cdot \text{Share}_b^{HP} \quad (3)$$

where

$$\text{Share}_b^{HP} = \frac{\sum_o \sum_{s \in \mathcal{S}^{HP}} OD_{o \rightarrow b, s}}{\sum_o \sum_{s \in \mathcal{S}} OD_{o \rightarrow b, s}} \quad (4)$$

is the share of commuters to block b employed in sectors with high physical-presence requirements. Here, $OD_{o \rightarrow b, s}$ denotes the number of workers commuting from origin block o to workplace block b in industry sector s ; $\mathcal{S}$ is the set of all industry sectors; and $\mathcal{S}^{HP} \subseteq \mathcal{S}$ denotes sectors that are, on average, more tightly bound to on-site work.

Finally, I aggregate the block-level measure to the metropolitan level by taking a share-weighted

¹¹In contrast, less physically constrained occupations include producer services such as finance, insurance, and arts-related activities. Although this classification is not without limitations, as some physically constrained sectors such as accommodation, health, and education are also included in this category, it remains reasonable to treat this group as relatively less constrained than strictly on-site occupations.

average across workplace blocks:

$$wMEBC_c = \frac{\sum_{b \in \mathcal{B}_c} EBC_b(N_c) \cdot \text{Share}_b^{HP}}{\sum_{b \in \mathcal{B}_c} \text{Share}_b^{HP}} \quad (5)$$

As a result, the onsite-commuting constrained $MEBC$, $wMEBC$, is a time-invariant measure between 0 and 1, where higher values indicate that commuting flows for on-site workers are concentrated on a small number of critical road segments. Road segments that are crucial for connecting locations within the city, therefore, reflect the extent to which city operations may be disrupted by traffic accidents, road construction, or severe weather conditions (Mattsson et al. (2015) Mattsson et al. (2015)).

Figure 4: Highest $wMEBC$: Flagstaff, AZ

Spatial decomposition of commuter-weighted network centrality: Flagstaff, AZ

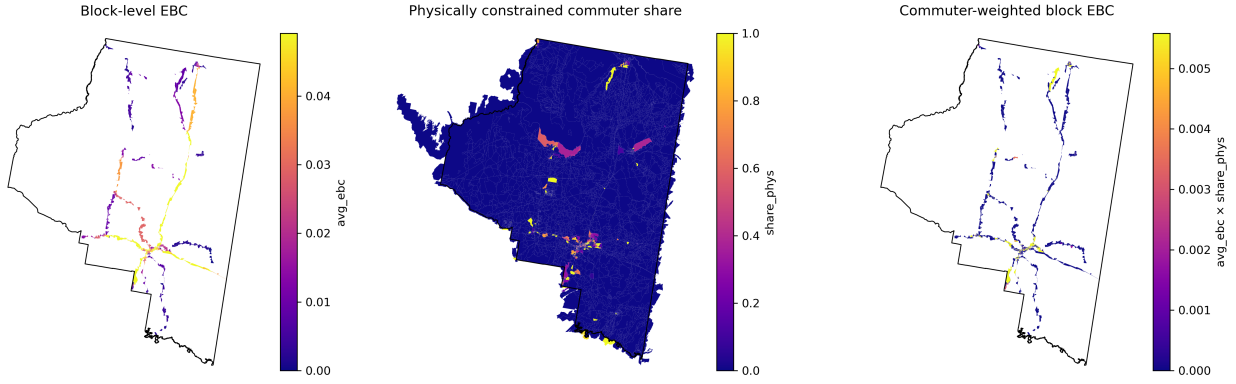


Figure 5: Lowest $wMEBC$: Houston-Sugar Land-Baytown, TX

Spatial decomposition of commuter-weighted network centrality: Houston-The Woodlands-Sugar Land, TX

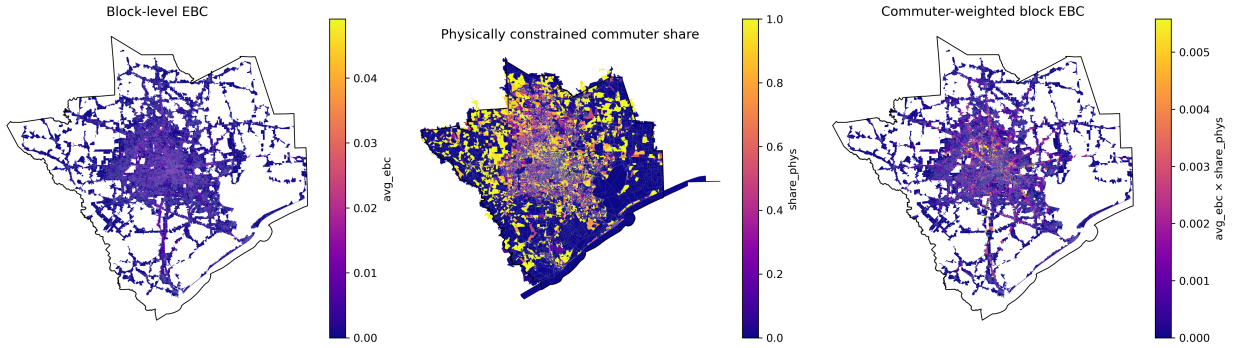


Figure 4 presents Flagstaff, Arizona, which has the highest $wMEBC$. First, Flagstaff's road network is highly constrained, with only a few primary corridors. As a result, a small number of

locations carry a disproportionately large share of shortest-path traffic. Along with this structural configuration, commuter flows in Flagstaff also display a similar pattern of concentration. With physically constrained commuting largely occurring along these same corridors rather than being dispersed across the network. The alignment between network centrality and commuter flows leads to a pronounced amplification effect in $wMEBC$. As a result, high $wMEBC$ values are spatially concentrated, indicating that the system depends heavily on a limited set of critical links, particularly for workers constrained to on-site locations.

In contrast, Figure 5 presents the MSA with the lowest $wMEBC$, Houston–Sugar Land–Baytown, TX. This region exhibits a highly distributed road network structure, indicating strong connectivity and the presence of numerous alternative routes. This configuration produces a relatively diffuse pattern of block-level betweenness centrality, with values distributed across many locations rather than concentrated in a few key nodes. In addition, commuter flows are similarly dispersed, with physically constrained commuting occurring across a broad spatial extent of the area. Because commuter activity is not strongly aligned with a limited set of high-centrality corridors, the interaction between network structure and commuting patterns does not generate strong localized amplification. Consequently, $wMEBC$ remains relatively low and spatially diffuse.

Intra-city Goods Trade Flow Concentration

Lastly, the flow of supplied goods plays a pivotal role in sustaining a city’s economic activity. I construct a measure of goods flow concentration, the intra-city traded flow ratio ($IntraTFR$), which captures the relative importance of internally sourced trade linkages. Specifically, $IntraTFR$ reflects the extent to which goods delivered to city c are sourced from within the same city, relative to all metro-origin shipments delivered to that city. The Freight Analysis Framework Version 5 (FAF5) Experimental County-Level Estimates, provided by the Bureau of Transportation Statistics, offer freight flow information by transportation mode and commodity group. While FAF5 reports shipment estimates at a coarser geographic level, an experimental county-level disaggregation factor, available only for 2022, is used to approximate county-to-county truck flows. To construct the pre-period $IntraTFR$ measure, I assume that the 2022 county-level disaggregation factors are unchanged from 2017. Applying this disaggregation factor, I first construct county-pair shipment volumes and then aggregate them to the MSA-pair level. In this step, I take the average shipment volume across all county pairs within each origin-MSA, destination-MSA, and commodity-group cell, capturing the average intensity of trade linkages between MSAs. I then aggregate these MSA-pair

Table 3: Interpretation of Agglomeration Forces

Measure	Range	Interpretation (higher values)
Labor Market (<i>HHI</i>)	[0, 1]	Fewer industries account for most jobs in the city
Road Network (<i>wMEBC</i>)	[0, 1]	A few key road segments carry most connections across locations, especially for on-site workers
Trade Flow (<i>IntraTFR</i>)	[0, 1]	More goods are sourced within the city

trade intensities across commodity groups to construct a city-level measure.

Based on this, the Intra-city Traded Flow Ratio (*IntraTFR*) is defined as follows:

$$IntraTFR_{c,y} = \frac{\sum_{j=c} T_{j \rightarrow c,y}}{\sum_j T_{j \rightarrow c,y}} \quad (6)$$

where $T_{j \rightarrow c,y}$ denotes the MSA-to-MSA shipment intensity from origin MSA j to destination MSA c in year y , constructed from average county-to-county shipment volumes within each MSA pair and aggregated across commodity groups. Accordingly, $\sum_{j=c} T_{j \rightarrow c,y}$ captures the inward trade intensity sourced from within MSA c , while $\sum_j T_{j \rightarrow c,y}$ captures the total inward trade intensity from all metro origins.

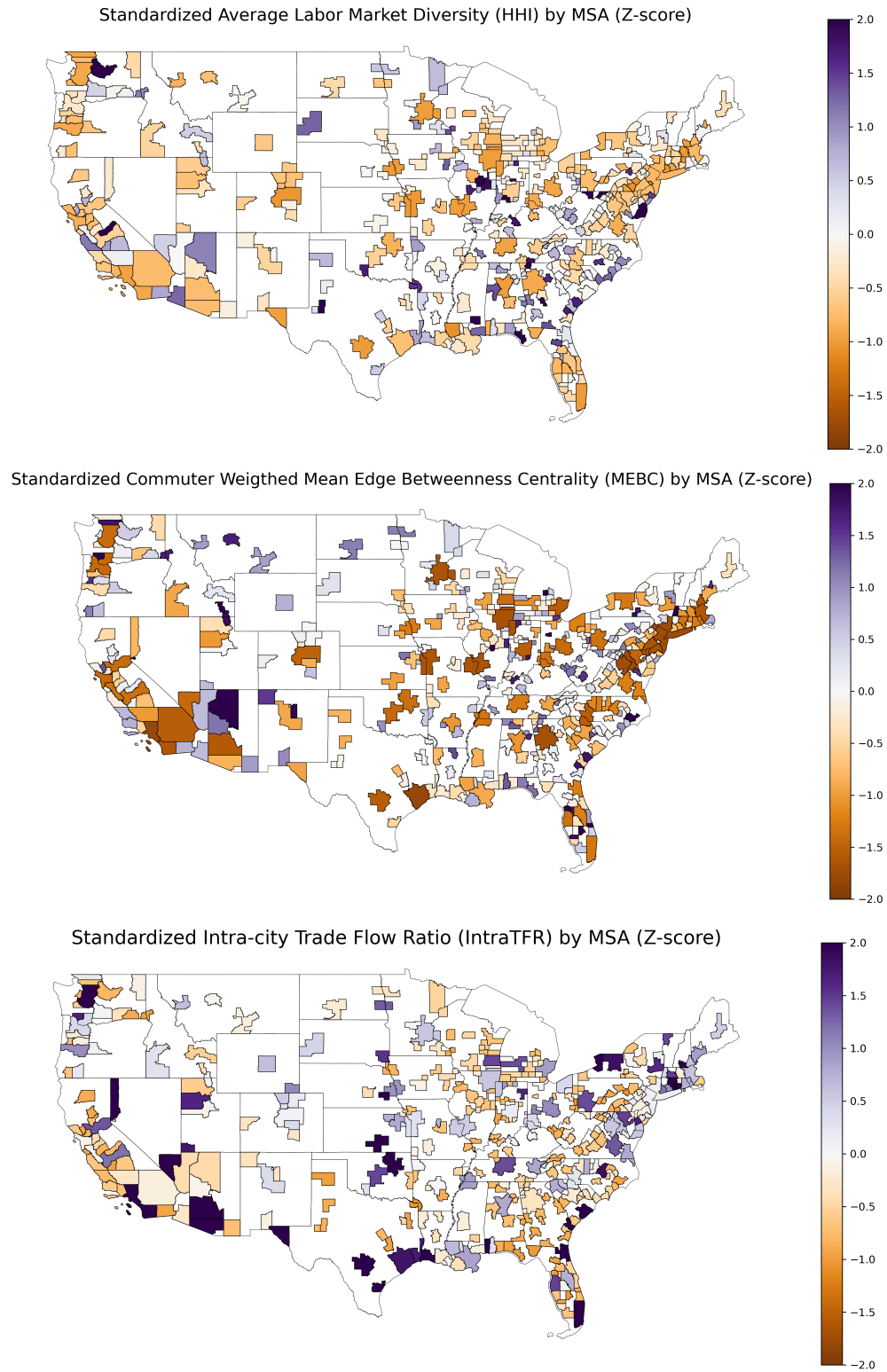
II.IV Correlation Across Three Dimensions of Urban Economic Structure

Three dimensions of urban economic structure, namely industrial composition, road network structure, and intra-urban trade linkages, capture distinct sources of heterogeneity in urban economic organization beyond the overall agglomeration intensity measured by population density. Figure 6 presents the correlations among the corresponding measures, the *HHI*, *wMEBC* and *IntraTFR*. They are only weakly correlated, and none exhibits a high degree of overlap with the others, suggesting that each captures a different aspect of a city’s economic structure. The strongest relationship is a moderate negative correlation of -0.46 between road network centrality and intra-city goods sourcing. This suggests that cities with more centralized road networks tend to rely less on intra-city goods sourcing, although the relationship remains moderate in magnitude. Figure 7 presents the MSA-level standardized pre-period measures of city economic structures, enabling direct comparisons across cities. The raw measures are shown in Appendix Figure 18. The standardized geographic distributions of these measures show substantial heterogeneity across MSAs.

Figure 6: Correlation Matrix of Agglomeration Measures at the MSA Level



Figure 7: City-Level Concentration Measures (2017): Standardized Z-Scores



Note: This figure presents the standardized Z-scores of average concentration measures for the labor market, road network, and trade perspectives across MSAs from 2017, allowing for cross-variable comparison.

III Research Design

III.I LP-DiD for Transient Natural Disasters

I begin by estimating how the economic effects of high-wind natural disasters vary with urban concentration characteristics using monthly variation in disaster exposure across MSAs. In this context, the exogenous timing and location of disaster events provide plausibly exogenous variation for identifying the causal effects of disaster exposure and their interaction with urban concentration characteristics. However, natural disasters occur as isolated, short-lived events unlike long-lasting policy interventions. Because MSAs may experience disasters repeatedly over time, treatment status can change from month to month, making it difficult to define sustained treatment and untreated control groups. Because MSAs may experience multiple disasters over time, treatment status can change from month to month, making it difficult to define persistent treatment and untreated control groups. For example, a disaster occurring in one month may be followed by another event two months later, causing a previously treated MSA to be exposed again after only a one-month interval. Likewise, an MSA that initially serves as a control may subsequently experience a disaster, preventing it from remaining untreated throughout the analysis period. As a result, treatment and control groups evolve continuously over time, violating the persistent treatment structure underlying conventional difference-in-differences designs. Consequently, the classical two-way fixed effect specification may be biased because repeated disaster episodes generate overlapping treatment effects across cities and over time. Even staggered DiD estimators are not well-suited to this setting because they likewise assume that treatment, once received, persists over time, as is typical for policy interventions (De Chaisemartin et al. (2020); Callaway et al. (2021); Goodman-Bacon (2021); Sun et al. (2021); Borusyak et al. (2024)).

The LP-DiD method is also advantageous in that it naturally accounts for past trends in the electricity consumption. Netting out the pre-trend of electricity consumption amounts is necessary to identify precisely the exact magnitude of each interaction’s impact on short-run economic disruptions. Overall, the LP-DiD framework provides a credible and flexible approach for estimating the causal effects of storm-related events by accommodating episodic on and off treatment status directly and accounting for pre-existing trends in electricity consumption. Given the need to identify valid treated and control MSAs, I apply the Local Projection Difference-in-Differences (LP-DiD) method proposed by Dube et al. (2023) and Jordà et al. (2025). While the LP-DiD framework is designed to accommodate staggered treatment timing and dynamic treatment effects, it is particularly

well-suited to capturing the dynamics of short-lived, non-persistent natural disasters because it does not require treatment status to persist following the initial exposure. For instance, it accommodates cases where an MSA is hit in month m but not in the following month $m + 1$, allowing for shifts in treatment status. Specifically, under the LP-DiD approach, I construct valid samples by retaining each MSA first exposed to a disaster in month t that remains treated through the post period of interest, denoted by $t + h$. I then pair it with MSAs that have not experienced a disaster over the overlapping window from $t - 2$ to $t + h$ ¹². A city is considered validly treated only if it is newly hit by a natural disaster in month t . In other words, I exclude MSAs that were hit in the previous month but are no longer treated in the current month to ensure stable treatment classification and to isolate the causal effect of disaster exposure from transitional dynamics. For each post-disaster month h , I construct period-specific samples and restrict the estimation to MSAs that remain in the corresponding sample.

Even when employing LP-DiD to mitigate issues of contaminated controls, concerns can persist regarding whether the selected control MSAs are truly comparable to treated MSAs in the context of natural disasters. Table 4 compares pre-period wind exposure across treated and control cities. The results show that treated MSAs have higher pre-period maximum wind speed, while control observations have slightly higher pre-period average wind speed. However, neither difference is statistically significant, suggesting no systematic difference in baseline wind exposure across the two groups.

Table 4: Pre-period Hurricane Exposure in Treated and Control MSA-Month Observations

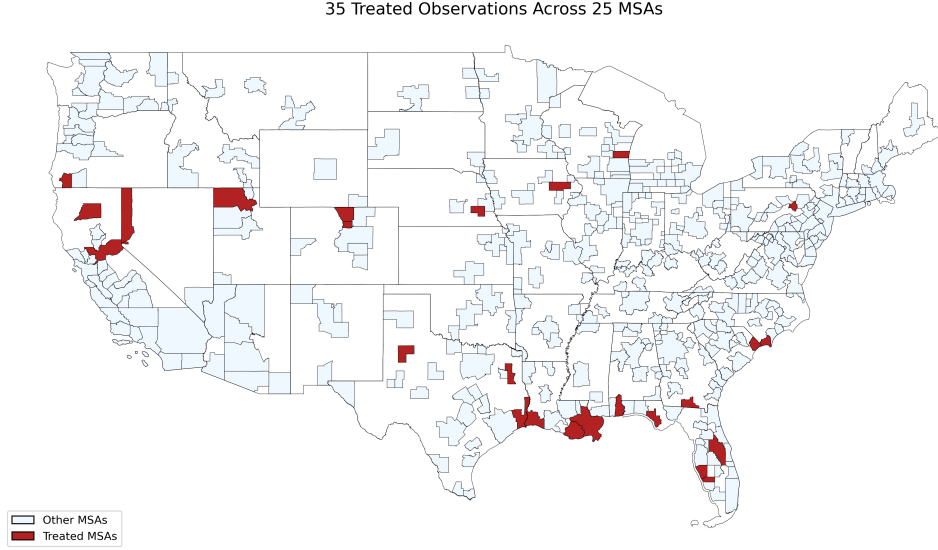
Variable	Control		Treated		Diff.	p-value
	Mean	SD	Mean	SD		
Pre-period Maximum Wind Speed	53.1535	23.3198	58.1143	29.4267	-4.9607	0.2086
Pre-period Average Wind Speed	19.6913	15.5623	16.8147	13.3796	2.8766	0.2744
Obs	30,467 MSA-months		35 MSA-months			
MSAs	365		25			

Note: This table reports pre-period hurricane exposure measures for the estimation sample at the MSA-month level. “Pre-period Maximum Wind Speed” is the maximum observed wind speed in a given MSA-month during the pre-period, while “Pre-period Average Wind Speed” is the corresponding average wind speed. The difference column reports Control minus Treated. P-values are from two-sample t-tests with equal variances.

¹²For control cities, I require no-hit status beginning in pre-event month when matching them with treated cities, assuming that the economic effects of a disaster generally stabilize within one month regardless of disaster intensity. As a robustness check, I extend the stabilization window from one to six months and find that the results remain highly consistent across specifications.

Figure 8 presents the geographic distribution of the 35 treated MSA-month observations across 24 unique treated MSAs identified under the LP-DiD framework over the 2018–2024 period. As expected, the treated MSAs are concentrated along coastal regions, particularly the Gulf Coast, although several inland MSAs are also included.

Figure 8: Geographic Distribution of Hit MSAs (2018-2024)



Note: The figure shows MSAs affected by natural disasters between 2018 and 2024 that experienced Category 2+ high-wind events (Saffir–Simpson scale). These MSAs are the treated units in the LP-DiD design.

III.II Econometric Framework

The adjusted estimating equation using the LP-DiD non-absorbing treatment framework by Dube et al. (2023) and Jordà et al. (2025), with a primary focus on the heterogeneous effects of disaster exposure based on city concentration measures, takes the following form.

$$\begin{aligned}
\ln(Load)_{c,m+h,y} - \ln(Load)_{c,m-1,y} &= (\delta_{m+h,y} - \delta_{m,y}) + \alpha_1 \Delta D_{c,m,y} \\
&\quad + \Delta D_{c,m,y} \left[\underbrace{\beta_1 HHI_c}_{\text{Labor Mkt: ind. comp.}} + \underbrace{\beta_2 wMEBC_c}_{\text{Road Net.: central.}} + \underbrace{\beta_3 IntraTFR_c}_{\text{Trade Flow: goods sourcing}} + \underbrace{\beta_4 PopDensity_c}_{\text{Urban form: density}} \right] \\
&\quad + \gamma_1 HDD_{c,m+h,y} + \gamma_2 CDD_{c,m+h,y} \\
&\quad + \gamma_3 OutageCount_{c,m+h,y} + \gamma_4 BuildingSh_{Pre-1980,c,y} \\
&\quad + \epsilon_{c,m+h,y}
\end{aligned} \tag{7}$$

In Equation 7, c denotes the MSA, m denotes month, h denotes post horizon with $h \geq 0$ and y denotes the calendar year from 2018 to 2024. Equation 7 excludes region fixed effects but instead includes month-by-year fixed effects following the LP-DiD method. The outcome variable, $\ln(\text{Load})_{c,m+h,y} - \ln(\text{Load})_{c,m-1,y}$ captures the change in logged industrial and commercial electricity consumption h periods after a disaster, relative to the consumption level in the month prior to the natural disaster hit. The variable $\Delta D_{c,m,y}$ equals one if MSA c is newly exposed in month m to high wind intensity equivalent to Saffir–Simpson Category 2 or above. Specifically, treatment is defined as one in the month an MSA is first struck by a high-intensity storm with maximum wind speeds exceeding 96 miles per hour. *PopDensity* is included to control for general city concentration intensity that may generate baseline differences in economic activities from the perspective of agglomeration forces. *HDD* and *CDD* represent heating and cooling degree days, respectively, measured at the MSA-month level, and are included to account for the influence of local weather conditions on electricity consumption. The terms $(\delta_{m+h,y} - \delta_{m,y})$ capture month-by-year fixed effects between the occurrence month and h periods ahead, absorbing aggregate time-varying factors common across metropolitan areas, including common seasonal fluctuations and nationwide shocks. Throughout the paper, standard errors are clustered at the MSA level to account for correlated shocks within each city, which may arise from disaster exposure patterns as well as from the structure of the sample. The coefficients of interest are from β_1 to β_3 which capture how disaster-driven impacts may be mitigated or amplified through each city’s agglomeration measures.

Table 5 presents summary statistics on electricity consumption, agglomeration measures, and climatological variables during the disaster-occurring month, comparing MSAs affected by disasters to unaffected MSAs serving as control groups in the LP-DiD framework.

Table 5: Summary Statistics for Treated and Control MSAs

Variable	Hit Cities		No-Hit Cities		Diff.	p-value
	Mean	SD	Mean	SD		
Log Electricity Sales	12.622	1.222	12.118	1.274	-0.504	0.019
Population Density	96.804	65.838	107.091	121.414	10.287	0.616
Labor Market Concentration	0.041	0.018	0.044	0.025	0.003	0.463
Road Network Centrality (Commuting Weighted)	0.013	0.007	0.014	0.006	0.001	0.491
Within-City Goods Sourcing Ratio	0.258	0.213	0.191	0.172	-0.067	0.022
<i>CDD</i> (thousands)	0.558	0.886	0.371	0.848	-0.187	0.193
<i>HDD</i> (thousands)	0.703	1.061	1.040	2.010	0.337	0.321
Obs	35		30,551			
MSAs	25		365			

Note: Summary statistics are reported at the onset month ($h = 0$), separately for MSAs newly exposed to high-wind disasters (Category 2 and above) and those not exposed. The table reports electricity sales, agglomeration measures, climate controls, and population density at the time of initial exposure. The difference column reports (Control minus Treated).

IV Results

IV.I Dynamic Responses to Natural Disaster Hits

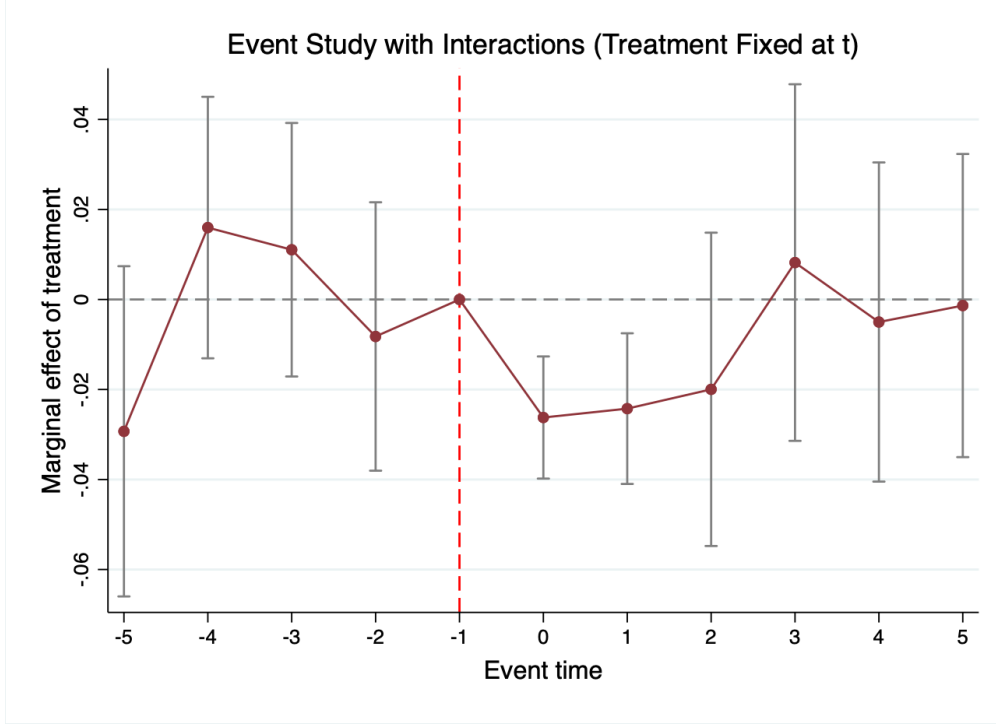
I begin by examining the dynamic response of electricity consumption to natural disaster hits. An important question is whether the associated economic disruption is concentrated in the event month or persists into subsequent periods. Tracing the response over time therefore provides evidence on both the immediate impact of a disaster and the persistence of its effects on electricity consumption. Figure 9 presents the corresponding event-study estimates. The analysis is constructed around isolated, one-period disaster shocks and uses clean-window controls. Specifically, I restrict the treatment sample to cities experiencing a natural disaster hit at (t) with no additional qualifying disaster hits over the subsequent five-month horizon¹³. Accordingly, the estimated coefficients should

¹³At the contemporaneous horizon ($(h = 0)$), the event-study sample coincides with the baseline LP-DiD non-absorbing specification. Differences arise only at post-treatment horizons ($(h > 0)$). The objective of the event study is to assess whether the effect of a disaster shock is concentrated in the event month or persists into subsequent periods. Accordingly, the dynamic specification is constructed around an isolated, one-period treatment shock rather than a sequence of continuously treated periods. This distinction matters because, under the standard implementation, estimating the treatment effect at horizon (h) requires treated observations to remain continuously treated through the corresponding post-treatment horizon. Such an estimand is less well suited to the present setting, where disasters are intermittent events and the object of interest is the response following a disaster occurring at (t). Requiring continuous treatment would also cause the number of eligible treated observations to decline rapidly as the event window expands. The event study therefore conditions on a single disaster occurrence and traces outcomes over the subsequent horizons. Control cities are required to remain untreated throughout the entire event window, spanning five months before to five months after the disaster hit.

be interpreted as the dynamic response to an isolated natural disaster hit rather than the effects of remaining continuously treated over multiple periods. In particular, the post-event coefficients trace the evolution of electricity consumption following a disaster month among treated cities that experience no additional qualifying disaster hits during the subsequent five-month event window.

Figure 9 indicates that the economic disruption induced by natural disaster exposure is highly transitory. The largest decline in electricity consumption occurs during the event month ($h = 0$), with economically meaningful effects persisting into the immediately following month ($h = 1$). Beyond one month, however, the estimated effects attenuate considerably and become statistically indistinguishable from zero. Taken together, the event-study evidence suggests that the economic consequences of natural disaster exposure are concentrated primarily in the event month, with limited persistence thereafter. This pattern motivates the focus of the subsequent analysis on the contemporaneous ($h = 0$) effect, which constitutes the primary estimand throughout the remainder of the paper.

Figure 9: Event-study with treatment assigned at the onset month only



Note: The figure reports event-study estimates of the effect of natural disaster hits on electricity consumption using isolated, one-period treatment shocks with clean-window controls. Event time is measured in months relative to the disaster hit ($h = 0$). The omitted category is the month immediately preceding the disaster hit ($h = -1$). At the contemporaneous horizon ($h = 0$), the estimation sample coincides with the baseline LP-DiD non-absorbing specification. For post-treatment horizons ($h > 0$), the event-study sample is constructed around isolated disaster events, with treated observations experiencing no subsequent disaster hits over the post-event window. Control cities remain untreated throughout the event window, spanning five months before to five months after the disaster hit. Vertical bars denote 95% confidence intervals based on standard errors clustered at the city level.

IV.II Baseline Results

Having established the dynamic pattern of the response to natural disaster hits, I now turn to the baseline estimates of their contemporaneous economic impact. The analysis focuses on Saffir–Simpson Category 2 or higher natural disasters and evaluates how these events affect local economic activity in the month of the disaster ($h = 0$). Unless otherwise stated, the results that follow therefore refer to the contemporaneous treatment effect. Table 6 reports the baseline estimates. Panel A presents the marginal treatment effect in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics. This provides a benchmark effect for a city with no particularly pronounced characteristics along any of the three urban-structure dimensions. The estimated coefficient of -0.0308 implies that, relative to the month preceding the disaster, monthly electricity load declines by approximately 3.1 percent in the month of occurrence.

Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. These coefficients characterize the direction in which the contemporaneous treatment effect varies across the three dimensions of urban structure, while the magnitude of this heterogeneity is examined separately below. All three interaction coefficients are statistically significant, indicating that the economic impact of natural disasters varies systematically with urban structure. Thus, the approximately 3 percent reduction in aggregate economic activity estimated for a city at the sample means of the three urban-structure characteristics masks substantial heterogeneity in disaster impacts across cities with different urban configurations. Greater industrial concentration, reflecting a larger share of employment concentrated in a limited number of industries, significantly amplifies disaster-related economic losses. In contrast, stronger road network connectivity and greater intra-city goods sourcing concentration mitigate these adverse effects.

Table 6: Contemporaneous Effects of Natural Disaster Exposure and Treatment-Effect Heterogeneity

		Change in Monthly MSA Electricity Load
		Event Month ($h = 0$)
<i>Panel A: Average Treatment Effect</i>		
	Average treatment effect	-0.0308^{***} (0.0061)
<i>Panel B: Interaction Coefficients</i>		
Labor Market	Industrial Composition (HHI)	-1.602^{***} (0.260)
	Road Connectivity ($wMEBC$)	2.964^{***} (0.958)
Trade Flow	Intra-city Goods Sourcing ($IntraTFR$)	0.0799^{***} (0.0307)
Demographics	Population Density	-0.000361^{***} (0.0000725)

Notes: Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics and other covariates. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in electricity load following natural disaster exposure, whereas negative coefficients indicate that they amplify it. Standard errors, clustered at the MSA level, are reported in parentheses. MSA-level electricity load is constructed using population-based weights. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To assess the economic significance of this heterogeneity, I next quantify how the estimated disaster effect changes across meaningful differences in each dimension of urban structure. Specifically, I evaluate the marginal treatment effect when each structural measure is increased or decreased by one standard deviation from its sample mean, while holding the other two dimensions at their respective sample means. This comparison can be interpreted as moving from the benchmark city

to one in which a given structural feature is meaningfully more or less pronounced, while the other dimensions of urban structure remain at their sample-average levels.

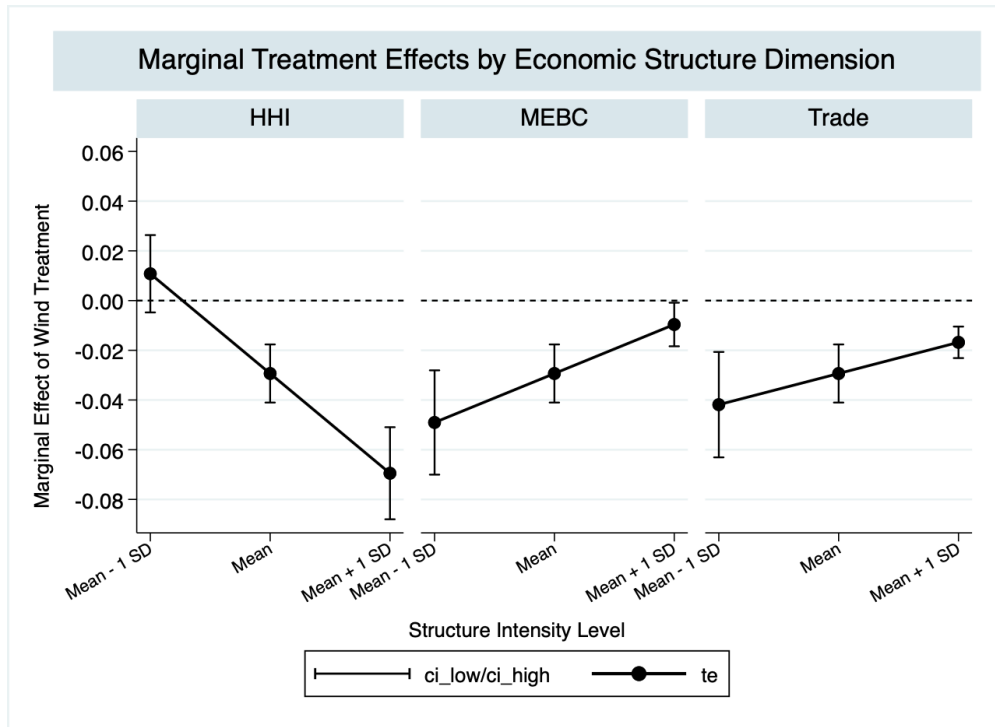
Table 7 and Figure 10 report the resulting treatment effects across these alternative urban configurations. The baseline effect corresponds to the benchmark MSA, with all three urban-structure characteristics evaluated at their sample means. The remaining estimates vary each characteristic individually by one standard deviation above and below its sample mean, while holding the other two characteristics at their sample means. These estimates provide a direct assessment of the economic importance of each urban-structure dimension in shaping local disaster resilience. For industrial composition, the heterogeneity is particularly pronounced. At one standard deviation above the mean of industrial concentration (HHI), a natural disaster is associated with an approximately 7 percent reduction in economic activity, compared with roughly 3 percent for the benchmark MSA. Thus, greater industrial concentration more than doubles the estimated economic loss from a disaster, with the treatment effect approximately 2.3 times as large as the benchmark effect. This represents the strongest gradient among the three dimensions of urban structure considered. In contrast, greater road connectivity substantially mitigates the adverse economic impact of natural disasters. A one-standard-deviation increase in network centrality ($wMEBC$) reduces the magnitude of the estimated treatment effect by approximately 40 percent relative to the benchmark, indicating that cities with greater concentration of commuting-weighted connectivity along critical road links experience smaller contemporaneous economic losses following a disaster. A similar pattern emerges for intra-city goods sourcing. A one-standard-deviation increase in sourcing concentration ($IntraTFR$) reduces the magnitude of the treatment effect by approximately 56 percent relative to the benchmark, suggesting that cities with more locally concentrated sourcing relationships are less adversely affected by disaster shocks. Together, these estimates reveal economically substantial heterogeneity: industrial concentration markedly increases disaster vulnerability, whereas the concentration of road connectivity along a few critical segments and greater localization of goods sourcing are associated with substantially smaller contemporaneous losses.

Table 7: Heterogeneous Treatment Effects by Urban Structure Intensity

		Treatment Effects Across Urban Structure Levels			Baseline Effect
		Monthly Electricity Load			
Category	Variable	Low	High	Difference	
Labor Market	Industrial Composition (<i>HHI</i>)	0.0093 (0.0076)	−0.0709*** (0.0100)	−0.0802*** (0.0130)	−0.0308*** (0.0061)
Road Network	Road Connectivity (<i>wMEBC</i>)	−0.0497*** (0.0111)	−0.0120** (0.0050)	0.0376*** (0.0122)	
Trade Flow	Intra-city Goods Sourcing (<i>IntraTFR</i>)	−0.0445*** (0.0108)	−0.0172*** (0.0035)	0.0274*** (0.0105)	
Observations: 30,499					

Notes: The Low and High columns report treatment effects evaluated at one standard deviation below and above the sample mean of each urban-structure measure, respectively. The Difference column reports the difference between the High and Low treatment effects. The Baseline Effect column reports the treatment effect evaluated with all three urban-structure measures fixed at their sample means. Standard errors, clustered at the MSA level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 10: Marginal Treatment Effects by Economic Structures



Note: Estimated marginal treatment effects of episodic natural disasters evaluated at the mean and one standard deviation above and below the mean of each economic structure intensity, holding all other covariates at their sample means. Error bars denote 95% confidence intervals.

V Industry-Specific Heterogeneity in Disaster Vulnerability

Results from Section IV.II show that various urban economic structures can either amplify or mitigate the economic impacts of natural disasters. Among the three dimensions considered, industrial concentration exhibits the strongest heterogeneity, with greater concentration substantially amplifying disaster-induced economic losses. This aggregate relationship, however, does not reveal which parts of the local industrial structure generate this vulnerability. Because the *HHI* summarizes concentration across all industrial subsectors, the estimated effect may reflect broad-based industrial concentration or instead be driven disproportionately by particular industry groups.

To distinguish between these possibilities, Section V decomposes overall industrial concentration into six industry group-specific *HHI* measures and examines their respective contributions to disaster vulnerability. The six industry groups comprise Retail, Utilities, and Construction (RUC), Health and Education (HE), Professional and Business Services (PS), Manufacturing, Accommodation, Transportation, and Wholesale Trade (ATT), and Other Industries. This extension also speaks to the well-established literature attributing recent urban growth primarily to the expansion and specialization of professional and business service industries, which have driven much of the recent productivity gains¹⁴ (Caliendo et al. (2018); Eckert et al. (2022); Rossi-Hansberg et al. (2021)). The analysis, therefore, asks whether the industries associated with contemporary agglomeration economies also play a disproportionate role in shaping vulnerability to natural disasters, or whether the aggregate relationship is driven by other parts of the local economy.

V.I Industry Group-Specific *HHI* Measures

To identify which parts of the local industrial structure contribute to the strong heterogeneity associated with overall industrial concentration, I decompose the baseline *HHI* into six industry group-specific components. For industry group g in MSA c , I define

$$HHI_{gc} = \sum_{i \in g} \left(\frac{Emp_{ic}}{TotalEmp_c} \right)^2, \quad g \in \{RUC, HE, PS, Manu, ATT, Other\} \quad (8)$$

¹⁴The professional and business services sector includes NAICS codes 51–56, comprising Information (51), Finance and Insurance (52), Real Estate and Rental and Leasing (53), Professional, Scientific, and Technical Services (54), Management of Companies and Enterprises (55), and Administrative and Support and Waste Management and Remediation Services (56). For instance, Eckert et al. (2022) show that between 1980 and 2015, urban-biased wage growth originated almost entirely within professional and business services, where the wage gap between high- and low-density areas widened substantially while remaining largely flat in other sectors.

where Emp_{ic} denotes employment in industry i and $TotalEmp_c$ denotes total employment in MSA c . The summation is restricted to industries belonging to group g , while employment shares continue to be defined relative to total MSA employment. Thus, HHI_{gc} captures the contribution of industries within group g to the MSA’s overall industrial concentration, rather than concentration within the group conditional on its own employment.

Because the six industry groups partition the industries entering the baseline concentration measure, the overall HHI can be written as the sum of these group-specific components:

$$\begin{aligned} HHI_c &= \sum_g HHI_{gc} \\ &= HHI_{RUC,c} + HHI_{HE,c} + HHI_{PS,c} + HHI_{Manu,c} + HHI_{ATT,c} + HHI_{Other,c} \end{aligned} \quad (9)$$

V.II Profiles of Industry Group Concentration Across Treated MSAs

I begin by characterizing the industry-group concentration profiles of treated MSAs across the six components defined above. These profiles show how different industry groups contribute to overall industrial concentration across disaster-exposed cities. The measures are constructed from each MSA’s average industry composition over the pre-treatment period from 2013 to 2017, thereby capturing persistent differences in local industrial structure rather than contemporaneous responses to disaster exposure.

Table 8 reports the distribution of treated MSAs across high, average, and low concentration categories for each industry group. The industry-group-specific HHIs are standardized across all MSAs and classified into quartiles, with the top quartile labeled “High”, the middle two quartiles “Average”, and the bottom quartile “Low”. It reveals substantial heterogeneity in industry-group concentration. First, the degree of concentration differs considerably across industry groups within the same MSA, indicating that overall industrial concentration masks important differences in the composition of specialization. Second, several MSAs exhibit high concentration in multiple industry groups simultaneously rather than in a single sector. For example, Ogden–Clearfield, UT is highly concentrated in RUC, HE, PS, and Manufacturing, while Beaumont–Port Arthur, TX exhibits high concentration in RUC, HE, Manufacturing, and ATT. These patterns underscore the multidimensional nature of local industrial specialization and the importance of distinguishing among its sectoral components when assessing heterogeneity in disaster impacts. Figure 11 provides a more concise visualization of these profiles by displaying, for each of the 25 treated MSAs, only the industry groups classified

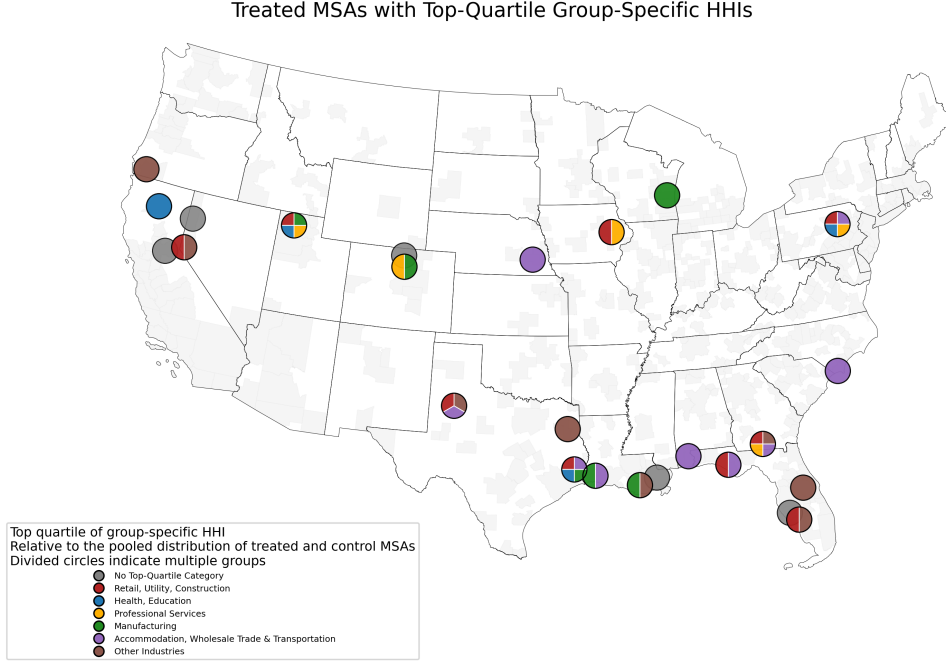
as highly concentrated.

Table 8: Industry-group concentration profiles of treated MSAs

MSA	RUC	HE	PS	Manu	ATT	Others
Beaumont-Port Arthur, TX	High	High	Low	High	High	Low
Bloomsburg-Berwick, PA	High	High	High	Avg	High	Avg
Boulder, CO	Low	Avg	High	High	Avg	Avg
Carson City, NV	High	Avg	Avg	Avg	Avg	High
Cedar Rapids, IA	High	Avg	High	Avg	Avg	Low
Daphne-Fairhope-Foley, AL	Avg	Low	Low	Avg	High	Avg
Fort Collins, CO	Avg	Low	Avg	Avg	Avg	Avg
Grants Pass, OR	Avg	Avg	Avg	Avg	Avg	High
Houma-Thibodaux, LA	Avg	Low	Avg	High	Avg	High
Lake Charles, LA	Avg	Avg	Avg	High	High	Avg
Lincoln, NE	Avg	Low	Avg	Avg	High	Avg
Longview, TX	Avg	Avg	Avg	Avg	Avg	High
Lubbock, TX	High	Low	Avg	Avg	High	High
Myrtle Beach-Conway-North Myrtle Beach, SC-NC	Avg	Low	Avg	Low	High	Low
New Orleans-Metairie, LA	Low	Avg	Avg	Avg	Avg	Avg
North Port-Sarasota-Bradenton, FL	Avg	Avg	Avg	Avg	Avg	Avg
Ogden-Clearfield, UT	High	High	High	High	Low	Avg
Orlando-Kissimmee-Sanford, FL	Avg	Avg	Avg	Low	Avg	High
Panama City, FL	High	Avg	Avg	Avg	High	Low
Punta Gorda, FL	High	Avg	Low	Low	Avg	High
Redding, CA	Avg	High	Avg	Avg	Avg	Avg
Reno, NV	Avg	Avg	Avg	Avg	Avg	Avg
Sacramento-Roseville-Arden-Arcade, CA	Avg	Avg	Avg	Low	Avg	Avg
Sheboygan, WI	Avg	Avg	Avg	High	Avg	Avg
Valdosta, GA	High	Avg	High	Avg	High	High

Note: **High**, Avg, and **Low** denote the upper, middle, and lower quartiles of the standardized industry-group concentration measures, respectively. The measures are standardized across all MSAs using average industry composition over the 2013–2017 pre-treatment period.

Figure 11: Geographic Distribution of Industry-Group Specialization Across Treated MSAs



V.III Results Using Industry Group-Specific HHI Measures

Building on the substantial variation in industry-group concentration documented across treated MSAs, I extend the baseline specification as shown in Equation 10, relaxing the restriction that concentration across different industry groups has a common effect on disaster impacts. Whereas the baseline specification summarizes industrial concentration using the aggregate measure HHI_c , the extended specification enters each industry group-specific component, HHI_{gc} , separately and allows it to have its own interaction coefficient. This decomposition therefore reveals whether the amplification associated with overall industrial concentration is broadly shared across sectors or concentrated in particular parts of the local economy. Estimating Equation 10 consequently allows me to identify which industry groups contribute most strongly to the heightened economic losses associated with greater industrial concentration.

$$\begin{aligned}
\ln(Load)_{c,m+h,y} - \ln(Load)_{c,m-1,y} = & (\delta_{m+h,y} - \delta_{m,y}) + \alpha_1 \Delta D_{c,m,y} \\
& + \Delta D_{c,m,y} \times \left[\beta_{11} HHI_{RUC,c} + \beta_{12} HHI_{HE,c} \right. \\
& \quad + \beta_{13} HHI_{BS,c} + \beta_{14} HHI_{MANU,c} \\
& \quad \left. + \beta_{15} HHI_{ATT,c} + \beta_{16} HHI_{OTHER,c} \right] \\
& + \Delta D_{c,m,y} \times \left[\beta_2 wMEBC_c + \beta_3 IntraTFR_c + \beta_4 PopDensity_c \right] \\
& + \gamma_1 HDD_{c,m+h,y} + \gamma_2 CDD_{c,m+h,y} \\
& + \gamma_3 OutageCount_{c,m+h,y} + \gamma_4 BuildingSh_{Pre-1980,c,y} \\
& + \epsilon_{c,m+h,y}
\end{aligned} \tag{10}$$

Table 9 reports the estimation results from Equation 10, which replaces the overall industry concentration measure with the full set of industry group-specific HHI measures. The results reveal substantial heterogeneity in the contributions of different industry groups to the aggregate effect of industrial concentration. In particular, concentration in Manufacturing appears to mitigate the economic consequences of natural disasters, although the estimated effect is only marginally statistically significant. In contrast, concentration in the HE and ATT industry groups exacerbates the economic consequences of natural disasters, with the estimated effect for ATT being highly statistically significant. Overall, these results demonstrate that the effect of industrial concentration is not homogeneous across industry groups. Rather, the aggregate exacerbating effect identified in the baseline specification masks substantial sectoral heterogeneity. Concentration in HE and ATT amplifies the economic consequences of natural disasters, whereas concentration in Manufacturing mitigates them.

Having established the heterogeneous roles of industry group-specific concentration in shaping the overall effect of industry concentration, Table 10 and Figure 12 illustrate how the estimated treatment effect varies over the observed range of all industry group-specific HHI measures (HHI_{gc}). Consistent with the regression results, higher concentration in the ATT and HE industry groups increasingly exacerbates the economic consequences of natural disasters, whereas greater concentration in the manufacturing industry increasingly mitigates them. Specifically, increasing manufacturing concentration from one standard deviation below the mean to the mean reduces the estimated treatment effect by roughly two-thirds. At the same time, increasing concentration in the

Table 9: Marginal Effects in the Event Month ($h = 0$): Industry Group-Specific HHI Measures

		Change in Monthly MSA Electricity Load
		Event Month ($h = 0$)
<i>Panel A: Average Treatment Effect</i>		
	Average treatment effect	−0.0192*** (0.00642)
<i>Panel B: Interaction Coefficients</i>		
Labor Market	Retail, Utilities, and Construction (HHI_{RUC})	−2.320 (3.335)
	Health and Education (HHI_{HE})	−2.812* (1.705)
	Professional Services (HHI_{PS})	0.710 (1.830)
	Manufacturing (HHI_{Manu})	2.479* (1.402)
	Accommodation, Transportation, and Trade (HHI_{ATT})	−1.531*** (0.322)
	Other Industries (HHI_{Other})	−0.097 (3.027)
Road Network	Road Connectivity ($wMEBC$)	2.830* (1.457)
Trade Flow	Intra-city Goods Sourcing ($IntraTFR$)	0.0762** (0.0297)
Demographics	Population Density	−0.000346*** (0.0000630)
Observations		30,499

Notes: Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all industry group-specific concentration measures and other urban structures. Panel B reports the estimated interaction coefficients between natural disaster exposure and each industry group-specific concentration measure, together with the remaining urban-structure characteristics. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in electricity load following natural disaster exposure, whereas negative coefficients indicate that they amplify it. Standard errors, clustered at the MSA level, are reported in parentheses. MSA-level electricity load is constructed using population-based weights. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

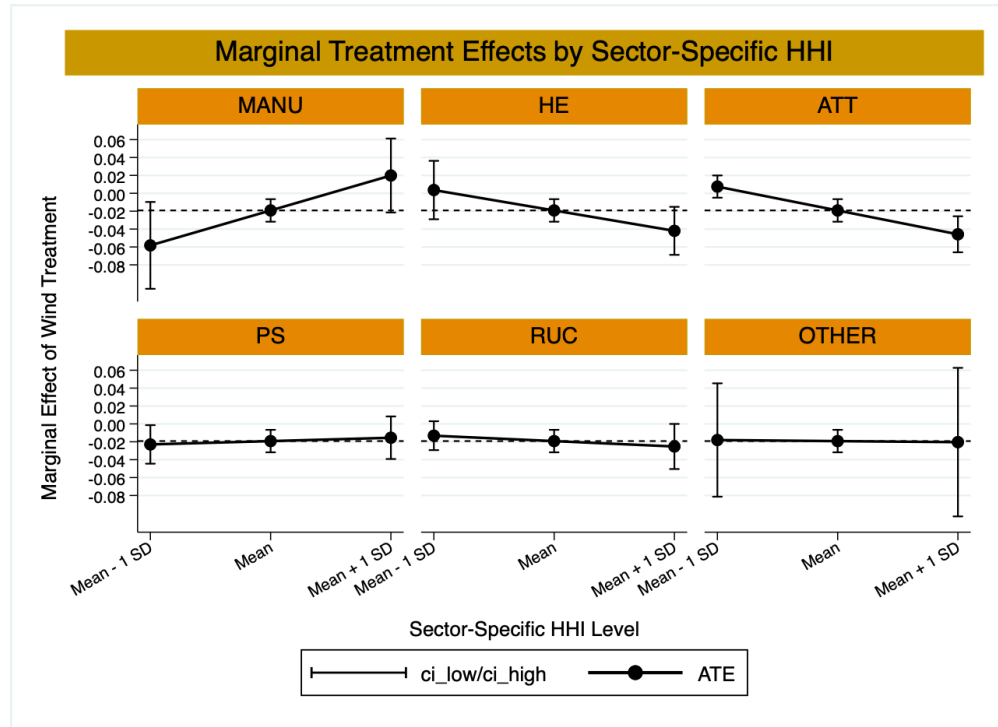
HE and ATT industry groups from the mean to one standard deviation above the mean increases the estimated decline in economic activity following natural disasters from approximately 2 percent to 4.2 percent and 4.6 percent, respectively. In contrast, the estimated treatment effects exhibit little variation across the RUC, PS, and Other industry groups.

Table 10: Heterogeneous Treatment Effects by Industry Group-Specific Concentration

Variable	Treatment Effects Across Industry Group Concentration			Baseline Effect
	Monthly Electricity Load			
	Low	High	Difference	
Manufacturing (HHI_{MANU})	-0.0582** (0.0248)	0.0199 (0.0211)	0.0781* (0.0442)	-0.0192*** (0.0064)
Health & Education (HHI_{HE})	0.0036 (0.0167)	-0.0420*** (0.0137)	-0.0456* (0.0276)	
Accommodation, Wholesale Trade & Transportation (HHI_{ATT})	0.0075 (0.0063)	-0.0459*** (0.0103)	-0.0533*** (0.0112)	
Producer Services (HHI_{PS})	-0.0229** (0.0110)	-0.0154 (0.0121)	0.0075 (0.0193)	
Retail, Utility, Construction (HHI_{RUC})	-0.0131 (0.0082)	-0.0252** (0.0129)	-0.0121 (0.0174)	
Other Industries (HHI_{OTHER})	-0.0180 (0.0323)	-0.0204 (0.0424)	-0.0024 (0.0743)	
Observations: 30,499				

Notes: The Low and High columns report treatment effects evaluated at one standard deviation below and above the sample mean of each industry group-specific concentration measure, respectively. The Difference column reports the change in the estimated treatment effect between the High and Low concentration levels. The Baseline Effect column reports the treatment effect evaluated with all industry group-specific concentration measures fixed at their sample means. Standard errors, clustered at the MSA level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 12: Marginal Treatment Effects by Industry Group-Specific



Note: Estimated marginal treatment effects of episodic natural disasters evaluated at the mean and one standard deviation above and below the mean of each industry group-specific HHI , holding all other covariates at their sample means. Error bars denote 95% confidence intervals.

V.IV Decomposing Industry Group-Specific HHI Measures

While the industry group-specific HHI , HHI_{gc} , are designed to capture concentration associated with each broad industry group, they incorporate two distinct features of local industry structure simultaneously: the relative scale of the industry group in the local economy and the degree to which employment within that group is specialized in a small number of subsectors. Because HHI_{gc} is constructed using total MSA employment as the denominator, variation in the measure can arise from either of these two dimensions. Accordingly, each group-specific HHI jointly captures the following two components.

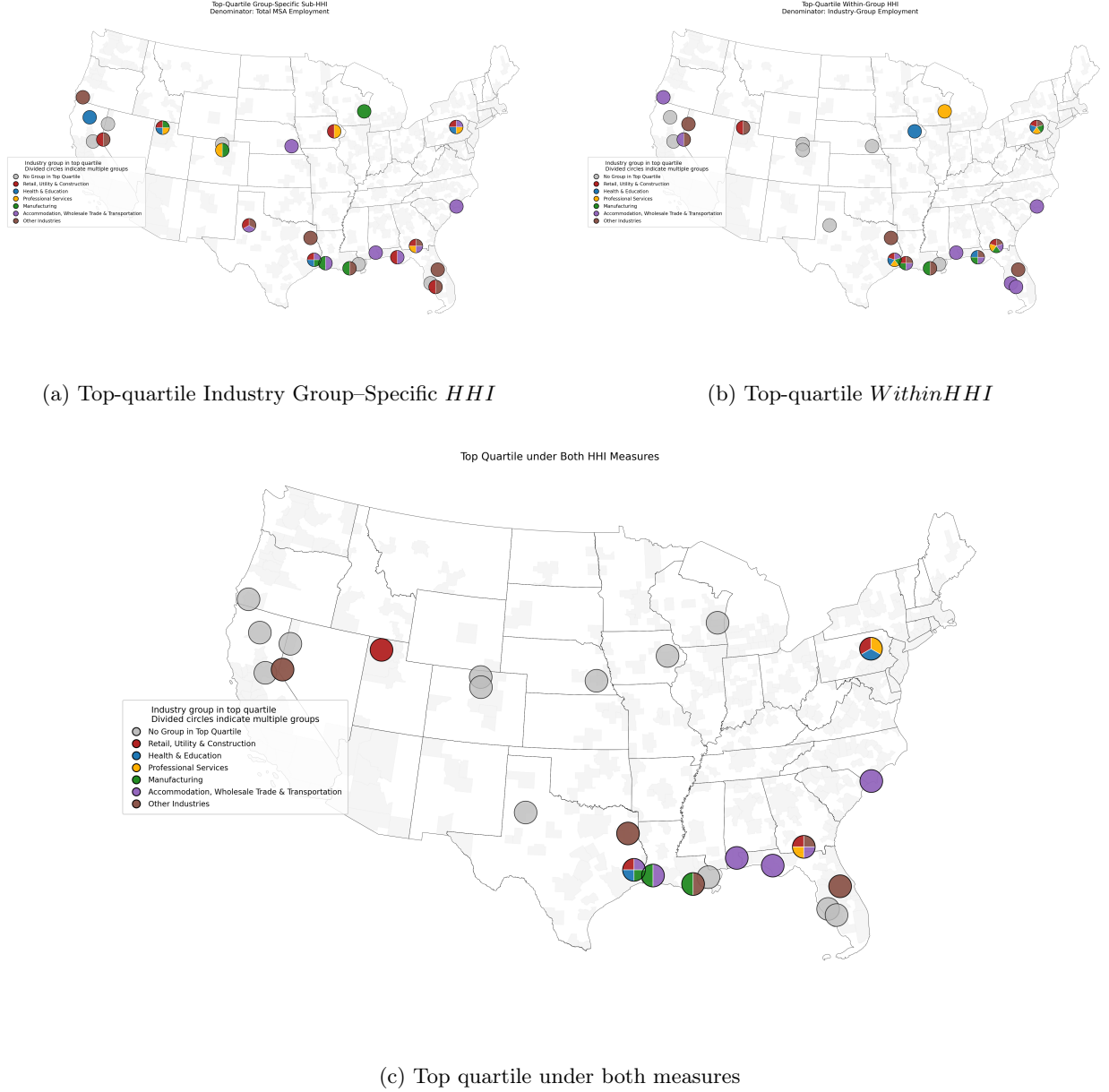
$$\begin{aligned}
HHI_{gc} &= \sum_{i \in g} \left(\frac{Emp_{ic}}{TotalEmp_c} \right)^2, \quad g \in \{RUC, HE, PS, Manu, ATT, Other\} \\
&= \sum_{i \in g} \left(\left(\frac{Emp_{ic}}{Emp_{gc}} \right) \times \left(\frac{Emp_{gc}}{TotalEmp_c} \right) \right)^2 \\
&= \left(\frac{Emp_{gc}}{TotalEmp_c} \right)^2 \sum_{i \in g} \left(\frac{Emp_{ic}}{Emp_{gc}} \right)^2 \\
&= Share_{gc}^2 \times WithinHHI_{gc}
\end{aligned} \tag{11}$$

Equation 11 shows how HHI_{gc} can be decomposed into two multiplicative components. The first component, $Share_{gc}^2$, is the squared employment share of industry group g in MSA c , capturing the relative size of the sector within the local economy. The second component, $WithinHHI_{gc}$, measures the concentration of employment across industries within group g . Thus, the industry group-specific HHI , HHI_{gc} , jointly captures two dimensions of local industry structure, the scale of the industry group and the degree of within-group specialization. The scale dimension reflects the importance of industry group g in the local economy, as measured by its share of total MSA employment. The specialization dimension reflects the extent to which employment within the broad industry group is concentrated in a small number of constituent industries rather than more evenly distributed across them. In other words, given that the overall employment concentration of an MSA (HHI_c) can be expressed as the sum of industry group-specific concentration measures (HHI_{gc}), and that each HHI_{gc} is, by construction, the product of the industry's share of total city employment and its within-group concentration ($WithinHHI_{gc}$), each industry's contribution to aggregate employment concentration from Section V.III reflects both the group's economic importance in the local economy

and the degree of concentration among its constituent industries.

Figure 13 compares the geographic distribution of treated cities classified in the top quartile under the HHI_{gc} and $WithinHHI$ measures. By removing the industry-size component from the concentration measure, the figure illustrates how the spatial distribution of highly concentrated cities changes when concentration is defined solely by the distribution of employment across detailed industries within each broad industry group. Panel C highlights the overlap between the two measures, identifying cities that remain highly concentrated after accounting for industry-group size. Cities that appear only under the industry group-specific HHI derive their high-concentration classification from the combination of industry-group scale and within-group specialization, whereas cities that remain highly concentrated under $WithinHHI$ exhibit substantial specialization within the industry group itself, independent of its relative size in the local economy. This decomposition allows me to investigate whether the observed heterogeneity across industry groups in shaping natural disaster impacts is driven primarily by differences in sectoral scale or by differences in within-group specialization.

Figure 13: Geographic Comparison of High Concentration Across Different HHI Measures



Note: Colored circles indicate industry groups in the top quartile. Divided circles indicate multiple industry groups in the top quartile within the same MSA. Gray circles indicate that no industry group is in the top quartile.

V.V Results Using Within-Group HHI Measures

Given that the heterogeneous mitigating and exacerbating effects of Manufacturing, ATT, and HE documented in Section V.III may arise through two distinct dimensions, it is useful to distinguish whether these effects reflect the relative importance of each industry group in the local economy or the degree of specialization among the detailed industries within the group. To isolate the

latter dimension, I next replace the industry group-specific HHI_{gc} measures with $WithinHHI_{gc}$ measures, which capture the degree of employment concentration across detailed industries within each broad industry group independently of the group’s employment share in the local economy. Comparing the results across the two measures allows me to assess whether the heterogeneous effects across industry groups are primarily associated with sectoral scale or with within-group specialization.

Table 11 reports the results using $WithinHHI$ measures, thereby isolating the role of employment concentration across detailed industries within each broad industry group. The results show that HE is the only industry group that exhibits a statistically significant exacerbating effect, indicating that greater concentration of employment in a relatively small number of HE sub-industries itself amplifies the economic consequences of natural disasters. Disaster vulnerability therefore increases not only when HE represents a larger share of metropolitan employment, but also when employment within the sector is concentrated in relatively few detailed industries. For example, an MSA whose HE employment is dominated by a small number of activities, such as hospitals or universities, exhibits greater within-group specialization than an MSA with a more diversified mix of hospitals, outpatient care, nursing facilities, schools, universities, social services, and other HE activities. In contrast, the exacerbating effect of ATT reported in Table 9 disappears. This finding suggests that the vulnerability associated with ATT primarily reflects its overall importance within the metropolitan economy rather than the distribution of employment across detailed ATT industries. Cities with a larger ATT workforce are therefore more vulnerable to disaster-related disruptions, regardless of how employment is allocated across the sector’s constituent industries. Similarly, the mitigating effect previously observed for manufacturing with the industry group-specific HHI measure is no longer statistically significant when using $WithinHHI$. This result indicates that the buffering effect of manufacturing likewise reflects the sector’s overall importance within the local economy rather than the distribution of employment across detailed manufacturing industries. In other words, cities with a larger manufacturing workforce appear to be more resilient, irrespective of the sector’s internal composition.

Table 12 and Figure 14 illustrate how the estimated marginal treatment effect varies with the level of each $WithinHHI$, capturing differences in the extent to which employment within each broad industry group is concentrated in a small number of detailed industries. Consistent with the interaction estimates reported in Table 11, meaningful heterogeneity is observed only for the HE sector. Specifically, increasing within-group concentration in the HE sector from the sample mean

Table 11: Marginal Effects in the Event Month ($h = 0$): Within-Industry-Group HHI Measures

		Change in Monthly MSA Electricity Load
		Event Month ($h = 0$)
<i>Panel A: Average Treatment Effect</i>		
	Average treatment effect	−0.0198** (0.0100)
<i>Panel B: Interaction Coefficients</i>		
Labor Market	Retail, Utilities, and Construction (<i>WithinHHI_{RUC}</i>)	−0.0313 (0.351)
	Health and Education (<i>WithinHHI_{HE}</i>)	−0.370*** (0.107)
	Professional Services (<i>WithinHHI_{PS}</i>)	0.306 (0.226)
	Manufacturing (<i>WithinHHI_{Manu}</i>)	−0.0369 (0.115)
	Accommodation, Transportation, and Trade (<i>WithinHHI_{ATT}</i>)	−0.0460 (0.0709)
	Other Industries (<i>WithinHHI_{Other}</i>)	0.0447 (0.0835)
Road Network	Road Connectivity (<i>wMEBC</i>)	2.756 (1.872)
Trade Flow	Intra-city Goods Sourcing (<i>IntraTFR</i>)	0.0674 (0.0516)
Demographics	Population Density	−0.000219** (0.000112)
Observations		26,019

Notes: Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all within-industry-group concentration measures and other urban structures. Panel B reports the estimated interaction coefficients between natural disaster exposure and each within-industry-group concentration measure, together with the remaining urban-structure characteristics. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in electricity load following natural disaster exposure, whereas negative coefficients indicate that they amplify it. The within-industry-group concentration measures (*WithinHHI*) capture the degree of employment concentration across detailed industries within each broad industry group. Standard errors, clustered at the MSA level, are reported in parentheses. MSA-level electricity load is constructed using population-based weights. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

to one standard deviation above the mean increases the estimated decline in economic activities following a natural disaster from 1.98 percent to 4.79 percent, more than doubling the estimated treatment effect. For the remaining industry groups, the estimated treatment effects vary across the observed range of within-group concentration, but these differences are not statistically distinguishable from zero.

Overall, the mechanism through which aggregate industrial concentration exacerbates the impacts of natural disasters operates through the heterogeneous roles of broad industry groups. Specifically, the ATT and HE industry groups amplify disaster impacts, whereas the Manufacturing group

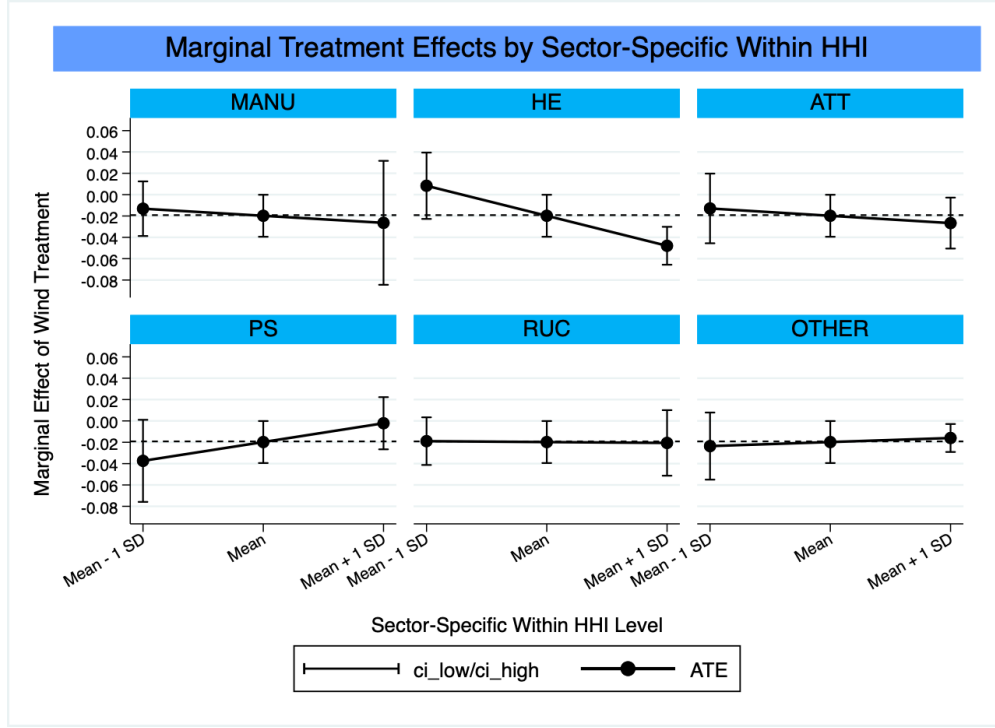
mitigates them. However, the mechanisms underlying these heterogeneous effects differ across sectors. For the ATT and Manufacturing groups, the heterogeneous effects are primarily driven by the sectors' relative scale and economic importance within the local economy, rather than by the degree to which employment is concentrated in a small number of detailed industries. In contrast, the exacerbating effect of the HE group persists even after abstracting from the sector-size component. This finding indicates that employment concentration within a relatively small number of HE sub-industries independently amplifies the impacts of natural disasters. Consequently, the vulnerability associated with the HE sector cannot be attributed solely to its importance in the local economy. Rather, internal specialization within the sector constitutes an additional and distinct source of disaster vulnerability.

Table 12: Heterogeneous Treatment Effects by Within-Industry-Group Concentration

Variable	Treatment Effects Across Within-Industry-Group Concentration			Baseline Effect
	Monthly Electricity Load			
	Low	High	Difference	
Manufacturing (<i>WithinHHI_{MANU}</i>)	−0.0132 (0.0131)	−0.0264 (0.0296)	−0.0132 (0.0411)	−0.0198** (0.0100)
Health & Education (<i>WithinHHI_{HE}</i>)	0.0083 (0.0159)	−0.0479*** (0.0091)	−0.0562*** (0.0162)	
Accommodation, Wholesale Trade & Transportation (<i>WithinHHI_{ATT}</i>)	−0.0129 (0.0166)	−0.0267** (0.0122)	−0.0137 (0.0212)	
Producer Services (<i>WithinHHI_{PS}</i>)	−0.0374* (0.0196)	−0.0022 (0.0125)	0.0352 (0.0260)	
Retail, Utility, Construction (<i>WithinHHI_{RUC}</i>)	−0.0190* (0.0114)	−0.0206 (0.0157)	−0.0017 (0.0186)	
Other Industries (<i>WithinHHI_{OTHER}</i>)	−0.0236 (0.0160)	−0.0160** (0.0067)	0.0076 (0.0142)	
Observations: 26,019				

Notes: The Low and High columns report treatment effects evaluated at one standard deviation below and above the sample mean of each within-industry-group concentration measure, respectively. The Difference column reports the change in the estimated treatment effect between the High and Low concentration levels. The Baseline Effect column reports the treatment effect evaluated with all within-industry-group concentration measures fixed at their sample means. Standard errors, clustered at the MSA level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 14: Marginal Treatment Effects by Industry Group-Specific



Estimated marginal treatment effects of episodic natural disasters evaluated at the mean and one standard deviation above and below the mean of each *withinHHI*, holding all other covariates at their sample means. Error bars denote 95% confidence intervals.

VI Road Network Centralization and Urban Resilience

VI.I Road Network Centralization and Mobility During Weather Shocks

The baseline results show that greater road network centralization yield substantially smaller contemporaneous economic losses following natural disasters. I next provide supporting evidence on a potential mechanism underlying this mitigating relationship by examining how road network structure shapes mobility during adverse weather conditions. If greater centralization facilitates the continuation of urban movement when transportation conditions are disrupted, cities with more centralized road networks should exhibit smaller reductions in mobility in response to weather shocks. Specifically, I explore whether more centralized road networks are correlated to improved mobility during weather-related disruptions. A natural implication of the proposed mechanism is that cities with more centralized road networks should be better able to maintain mobility during externally driven disruptions. In a highly centralized network, a relatively small number of road segments serve as the primary connectors across the city. As a result, prioritizing the restoration

and operation of these critical corridors may preserve network functionality more effectively than in decentralized networks, where connectivity is distributed across a larger number of routes. The concentration of essential links may therefore provide a structural basis for the mitigating effect of road network centralization documented in the baseline analysis. This interpretation is consistent with transportation recovery practices. Federal Highway Administration (2013) emphasize that post-disaster traffic management prioritizes restoring “essential traffic” along key transportation corridors to facilitate emergency response and accelerate network recovery. Because centralized road networks concentrate connectivity along a smaller set of strategically important links, prioritizing these corridors may restore a larger share of overall network functionality more rapidly, thereby limiting mobility disruptions and contributing to greater urban resilience.

To assess whether more centralized road networks are associated with smaller traffic disruptions during adverse weather conditions, I relate road network centralization to weather-related vehicle hours of delay (VHD), a measure of traffic delay attributable to adverse weather. Traffic performance measures are obtained from the Regional Integrated Transportation Information System (RITIS) Causes of Congestion database. The data are available at the county-month level for 2019 and report delays relative to free-flow travel conditions. RITIS attributes congestion to several primary causes, including recurrent congestion, traffic incidents, work zones, holidays, weather, traffic signals, and combinations of these factors. VHD measures the cumulative hours of delay experienced by vehicles relative to free-flow travel, where one VHD represents one vehicle experiencing one additional hour of delay. Especially, weather-related VHD captures congestion attributable to adverse weather conditions such as rain, snow, fog, and ice. Lower weather-related VHD therefore indicates that the transportation network is better able to maintain mobility during weather-induced disruptions.

I compare the partial correlations between $wMEBC$ and different categories of VHD, controlling for metropolitan population size. Controlling for population is important because larger MSAs tend to exhibit both lower $wMEBC$ and higher levels of congestion across all categories. Consequently, unconditional correlations largely capture differences in metropolitan scale and travel demand rather than the relationship between road network centralization and traffic performance. Indeed, without adjusting for population, all VHD categories exhibit similarly strong negative correlations with $wMEBC$ ¹⁵. After controlling for metropolitan population size, however, Table 13 and Figure 15 reveal substantial differences across congestion categories, allowing the relationship between road

¹⁵Appendix Table ?? reports the unconditional correlations. As expected, larger metropolitan areas exhibit both lower $wMEBC$ and higher congestion across all categories, making metropolitan population an important confounding factor.

network centralization and specific sources of traffic delay to be more clearly identified.

Table 13 shows that the partial correlation between weather-related VHD and $wMEBC$ is negative. This relationship indicates that more centralized road networks are associated with lower levels of weather-induced traffic delay, after controlling for city population size. In contrast, the partial correlation between recurrent congestion and $wMEBC$ is positive, consistent with the expectation that routine congestion is more severe in road networks where connectivity is concentrated along a limited number of major corridors. The contrasting relationships suggest that, although greater road network centralization may be associated with higher recurrent congestion under normal operating conditions, it may also help preserve mobility when transportation networks are exposed to weather-related disruptions. This contrast provides corroborating evidence for the baseline result that higher $wMEBC$ yields smaller economic losses following natural disasters and offers a plausible explanation for the greater resilience of more centralized road networks.

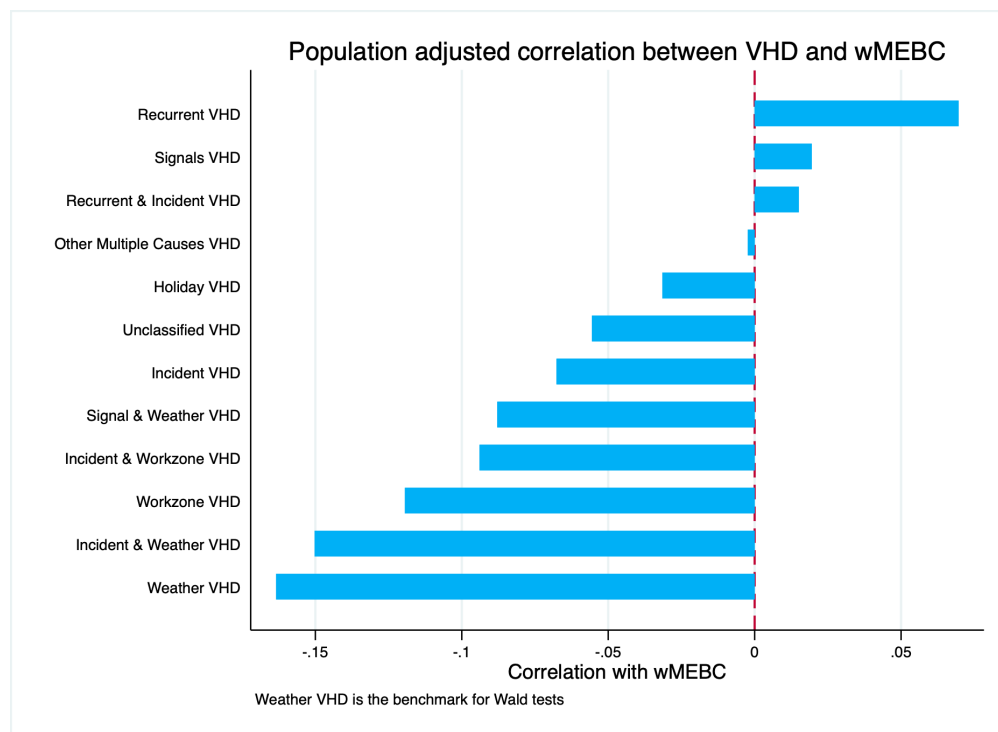
Table 13: Partial Correlations Between Weighted Mean Edge Betweenness Centrality and Vehicle Hours of Delay, Controlling for City Population

VHD Category	Partial Correlation	Difference from Weather
Weather VHD	-0.163	Reference
Incident & Weather VHD	-0.150	
Workzone VHD	-0.119	
Incident & Workzone VHD	-0.094	*
Signal & Weather VHD	-0.088	*
Incident VHD	-0.068	**
Unclassified VHD	-0.056	**
Holiday VHD	-0.032	***
Other Multiple Causes VHD	-0.002	***
Recurrent & Incident VHD	0.015	***
Signals VHD	0.020	**
Recurrent VHD	0.070	***

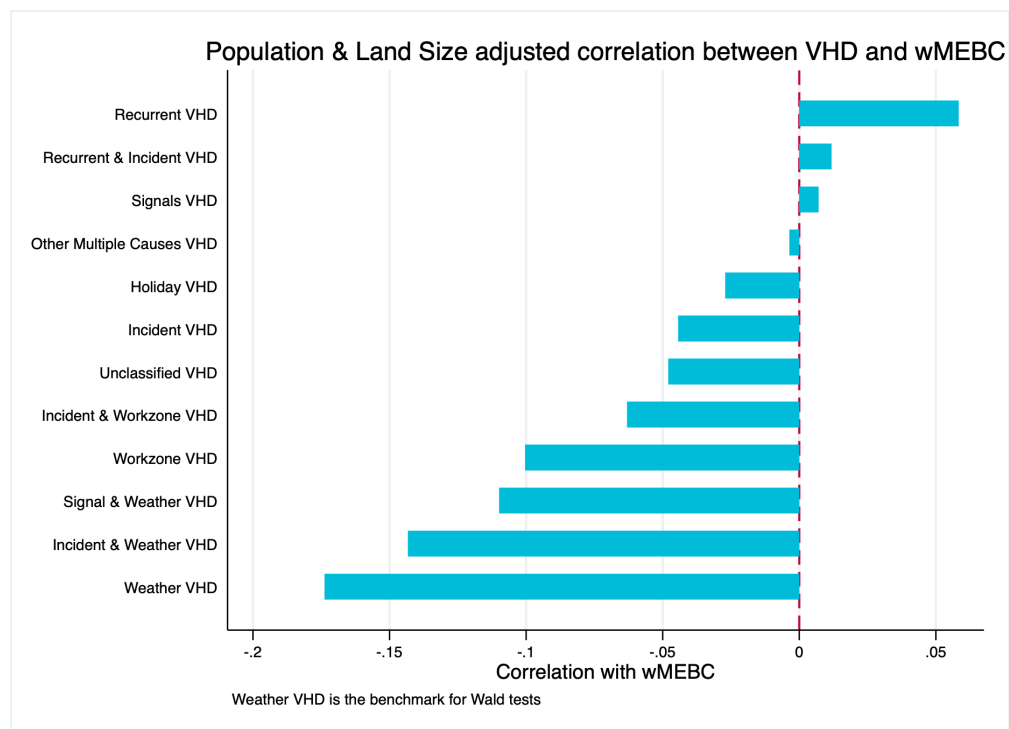
Notes: Entries report partial correlations between weighted Mean Effective Betweenness Centrality ($wMEBC$) and Vehicle Hours of Delay (VHD), estimated from a covariance structural equation model (SEM) controlling for city population. Weather VHD serves as the reference category. The “Difference from Weather” column reports the significance of Wald tests comparing each VHD- $wMEBC$ partial correlation with the Weather VHD- $wMEBC$ partial correlation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 15: Correlations between VHD categories and $wMEBC$ under alternative adjustment specifications

(a) Adjusted for population



(b) Adjusted for population and land area



Note: Bars represent Pearson correlation coefficients between each VHD category and $wMEBC$.

VII Alternative Measures of Economic Activity

As a broad monthly measure of metropolitan economic activity, the baseline analysis uses commercial and industrial electricity consumption. Because electricity demand responds immediately to business disruptions and reflects the intensity of commercial operations and the utilization of industrial and business capital, it provides a timely measure of firms' production activity following natural disasters. Firms, however, may adjust to disruptions over different time horizons and through multiple operational margins rather than production alone. To examine whether the main findings extend beyond the contemporaneous electricity-consumption response, I consider two additional measures of economic activity.

First, I replace monthly electricity consumption with its three-month moving average, which captures firms' average operating intensity over a broader time window while reducing the influence of transitory month-to-month fluctuations. This smoothing, however, comes at the cost of attenuating short-run variation, making the measure less sensitive to the immediate economic disruption caused by a natural disaster. Second, I examine the three-month average change in employment, which captures adjustment along the labor margin through the average pace of hiring and layoffs over recent months. Together, these measures allow me to assess whether the baseline patterns persist when economic activity is measured over a broader temporal window and through a distinct margin of firm adjustment. The following subsections consider each measure in turn.

VII.I Average Operating Intensity

I first examine firms' operating responses over a broader temporal window. Whereas the baseline outcome captures the immediate response of commercial and industrial electricity consumption in the disaster month, the three-month moving average captures the average intensity of business operations over recent months. Comparing the two measures therefore provides a useful distinction between the immediate disruption associated with a natural disaster and its effect on firms' average operating intensity over a broader period.

Table 14 reports the event-month estimates using the logarithm of three-month average commercial and industrial electricity consumption as the outcome variable, which captures firms' average operating intensity over a broader temporal window. Compared with the baseline estimate of firms' immediate operating response, measured using the logarithm of event-month commercial and industrial electricity consumption and reported in Table 6, the estimated marginal effect of natural

disaster exposure declines by approximately half, to 1.34 percent. This smaller effect is consistent with the three-month average smoothing the immediate disruption observed in the event month, resulting in a more moderate decline in firms' average operating intensity than in their contemporaneous operating intensity. Furthermore, the heterogeneous effects across urban economic structures differ from those obtained using monthly electricity consumption. Most notably, industrial concentration continues to play a significant role in shaping firms' average operating intensity following natural disasters. By contrast, the buffering effects of centralized road connectivity and intra-city goods sourcing become substantially weaker and are no longer statistically significant. These results suggest that road networks and local supplier linkages primarily mitigate immediate operational disruptions rather than shaping firms' average operating intensity following natural disasters. Industrial composition, however, remains an important determinant of firms' average operating intensity following natural disasters, even when operating activity is measured using a three-month average rather than monthly electricity consumption.

VII.II Labor Market Adjustments

The baseline analysis uses electricity consumption as a broad, high-frequency measure of economic activity to capture the immediate effects of natural disasters. In this section, the perspective shifts slightly to consider these responses through the lens of firms' adjustment decisions. In particular, industrial and commercial electricity consumption reflects firms' operating activity, so the preceding results can be interpreted as capturing firms' immediate adjustments in operating intensity following a disaster. Firms, however, may respond along other margins as well, including the labor margin through changes in hiring and layoffs. I therefore examine employment dynamics to assess whether differences in urban economic structure that shape firms' immediate operating responses also manifest in their labor market adjustments following natural disasters.

To capture these labor market adjustments, changes in employment provide a more direct measure of firms' contemporaneous hiring and layoff decisions in response to natural disasters than employment levels. Employment levels represent the cumulative outcome of past hiring and separation decisions and therefore tend to adjust more gradually. In the short run, firms may initially respond to a disaster by slowing or postponing planned hiring rather than immediately laying off existing workers, particularly if they expect normal operations to resume after a temporary disruption. As a result, the pace of employment growth may slow even while the overall level of employment remains relatively stable in the immediate aftermath of a disaster. I choose the three-month aver-

Table 14: Contemporaneous Effects of Natural Disaster Exposure on the Three-Month Average Commercial and Industrial Electricity Consumption

		Three-Month Average Log Electricity Consumption
		Event Month ($h = 0$)
<i>Panel A: Average Treatment Effect</i>		
Average treatment effect		−0.0134*** (0.00456)
<i>Panel B: Interaction Coefficients</i>		
Labor Market	Industrial Composition (HHI)	−0.741*** (0.235)
	Road Connectivity ($wMEBC$)	0.768 (0.495)
Trade Flow	Intra-city Goods Sourcing ($IntraTFR$)	0.00858 (0.0269)
Demographics	Population Density	−0.000208** (0.0000837)
Observations		25,263

Notes: Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics and other covariates. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in three-month average commercial and industrial electricity consumption following natural disaster exposure, whereas negative coefficients indicate that they amplify it.

The dependent variable is the logarithm of the three-month average commercial and industrial electricity consumption,

$$\ln\left(\frac{Load_{it} + Load_{i,t-1} + Load_{i,t-2}}{3}\right),$$

where $Load_{it}$ denotes commercial and industrial electricity consumption in MSA i during month t . Relative to the baseline specification based on monthly electricity consumption, this measure captures firms' average operating intensity while reducing the influence of transitory month-to-month fluctuations.

Treatment is defined by an eligible storm onset in the event month. The estimated coefficient of −0.0134 implies that, evaluated at the sample means of all urban-structure characteristics, natural disaster exposure reduces the three-month average commercial and industrial electricity consumption by approximately 1.34 percent relative to the corresponding counterfactual outcome.

Control MSAs are required to exhibit no treatment-status transitions during the relevant pre-event stabilization window. Standard errors, clustered at the MSA level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

age change in employment as the labor market outcome¹⁶¹⁷. The three-month average employment

¹⁶Specifically, I examine the three-month average change in employment. The three-month average change in employment for metropolitan area i in month t is defined as

$$\Delta Emp_{it}^{(3)} = \frac{1}{3} \sum_{k=0}^2 (Emp_{i,t-k} - Emp_{i,t-k-1}),$$

where Emp_{it} denotes CES employment (in thousands of employees). Positive values indicate employment expansion, while negative values indicate employment contraction.

¹⁷Regarding the use of three-month averages, because the CES–SAE is a sample-based establishment survey, one-month employment changes are subject to considerable sampling variability. Averaging employment changes over the preceding three months reduces transitory sampling noise and provides a more stable measure of the underlying pace

change therefore captures whether temporary disruptions to business operations translate into a slower pace of labor market adjustment, complementing commercial and industrial electricity consumption by measuring a distinct margin of firms' post-disaster response.

Table 15 reports the event-month results using the CES–SAE three-month average employment change to examine firms' adjustment along the labor margin. Natural disasters reduce the three-month average monthly employment change by approximately 574 workers relative to the corresponding pre-event period. This reduction is significantly smaller in MSAs with greater road connectivity, whereas it is significantly larger in MSAs with stronger intra-city trade networks, consistent with the baseline findings. Together, these results suggest that urban structure plays an important role in shaping firms' labor market adjustments following natural disasters. Importantly, the patterns of heterogeneity across urban structures closely mirror those documented for commercial and industrial electricity consumption, despite capturing a distinct margin of firms' adjustment. Then, Table 16 reports the estimated marginal treatment effects on the three-month average monthly employment change, measured in thousands of workers, evaluated across varying levels of each urban structural characteristic. Evaluated at the mean values of all urban structure dimensions, representing an MSA in which no particular dimension of urban structure is especially strong or weak relative to the sample, natural disasters reduce the three-month average monthly employment change by approximately 570 workers, as discussed previously. For instance, a one-standard-deviation increase in industrial concentration exacerbates the estimated reduction to approximately 945 workers, representing an effect that is about 65 percent larger than at the mean.

Taken together, these two outcome measures capture distinct margins through which firms adjust to natural disaster exposure. The three-month average of commercial and industrial electricity consumption captures firms' operating intensity, whereas employment changes capture adjustment along the labor margin, reflecting the net effect of firms' hiring and separation decisions. Despite these conceptual differences, the overall empirical patterns remain remarkably consistent. Industrial concentration continues to play the primary role in shaping firms' operating activity, remaining statistically significant even when operating intensity is measured using the three-month average of electricity consumption. By contrast, centralized road connectivity and intra-city goods sourcing remain important determinants of labor market adjustment.

of employment adjustment.

Table 15: Contemporaneous Effects of Natural Disaster Exposure on the Three-Month Average Employment Change

		Three-Month Average Employment Change
		Event Month ($h = 0$)
<i>Panel A: Average Treatment Effect</i>		
Average treatment effect		−0.574*** (0.220)
<i>Panel B: Interaction Coefficients</i>		
Labor Market	Industrial Composition (HHI)	−14.903 (11.286)
Road Network	Road Connectivity ($wMEBC$)	48.225** (19.525)
Trade Flow	Intra-city Goods Sourcing ($IntraTFR$)	1.551*** (0.564)
Demographics	Population Density	−0.00987*** (0.00310)
Observations		30,436

Notes: Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics and other covariates. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in the three-month average employment change following natural disaster exposure, whereas negative coefficients indicate that they amplify it.

The three-month average employment change is defined as

$$\overline{\Delta E}_{it}^{(3)} = \frac{E_{it} - E_{i,t-3}}{3},$$

where E_{it} denotes CES–SAE employment in MSA i and month t , measured in thousands of workers. The measure therefore represents the average monthly change in employment over the preceding three months and may take either positive or negative values.

Treatment is defined by an eligible storm onset in the event month. The estimated coefficient of −0.574 implies that natural disaster exposure is associated with an additional decline of approximately 0.574 thousand workers, or 574 workers, in the average monthly employment change relative to the corresponding counterfactual outcome.

Control MSAs are required to exhibit no treatment-status transitions during the relevant pre-event stabilization window. Standard errors, clustered at the MSA level, are reported in parentheses. Electricity-specific controls, including heating degree days, cooling degree days, outage measures, and the share of pre-1980 housing stock, are excluded. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 16: Heterogeneous Treatment Effects by Urban Structure Intensity: Average Three-Month Employment Changes (CES-SAE)

		Treatment Effects Across Urban Structure Levels: Event Month ($h = 0$)			
		Average Three-Month Employment Changes			Baseline Effect
Category	Variable	Low	High	Difference	
Labor Market	Industrial Composition (HHI)	-0.2038 (0.2872)	-0.9451** (0.4143)	-0.7413 (0.5614)	-0.5744*** (0.2197)
	Road Connectivity ($wMEBC$)	-0.8773*** (0.2900)	-0.2715 (0.2063)	0.6059** (0.2453)	
Trade Flow	Intra-city Goods Sourcing ($IntraTFR$)	-0.8446*** (0.2770)	-0.3042 (0.1977)	0.5404*** (0.1963)	
Observations: 30,436					

Note: Low and High columns report estimated treatment effects on average three-month employment changes (CES-SAE), evaluated at one standard deviation below and above the mean of each urban structure measure. The Difference column reports the difference between the High and Low treatment effects, while the Baseline column reports the treatment effect evaluated at the sample means. Because employment is measured in thousands of workers, the baseline estimate of (-0.5774) indicates that an ND-HIT month reduces the change in the average three-month employment count by approximately 0.58 thousand workers, or about 577 workers, relative to the change that would have occurred without an ND-HIT. Standard errors are clustered at the MSA level. Observations: 30,436.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VIII Robustness Checks

VIII.I Alternative Weighting Scheme for Electricity Consumption Distribution

In the baseline analysis, I allocate utility-level electricity load to MSAs using the population shares of the counties comprising each MSA to construct monthly commercial and industrial electricity consumption. As an alternative, I instead use CBP employment shares as the allocation weights. Figure 3 shows that the resulting MSA-level electricity consumption series are highly consistent across the two weighting schemes. Consistent with this finding, Table 17 shows that re-estimating the baseline specification using the employment-weighted electricity consumption measure yields results that are highly similar to the baseline estimates.

Table 17: Robustness to Alternative Electricity-Load Weighting Scheme

		Change in Monthly MSA Electricity Load
		CBP Employment Weights
<i>Panel A: Average Treatment Effect</i>		
	Average treatment effect	−0.0306*** (0.0061)
<i>Panel B: Interaction Coefficients</i>		
Labor Market	Industrial Composition (<i>HHI</i>)	−1.656*** (0.276)
	Road Connectivity (<i>wMEBC</i>)	2.875*** (0.953)
Trade Flow	Intra-city Goods Sourcing (<i>IntraTFR</i>)	0.0747** (0.0301)
Demographics	Population Density	−0.000372*** (0.0000726)
Observations		26,019

Notes: Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics and other covariates. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. MSA-level electricity load is constructed using County Business Patterns (CBP) employment-based weights rather than the population-based weights used in the baseline specification. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in electricity load following natural disaster exposure, whereas negative coefficients indicate that they amplify it. Standard errors, clustered at the MSA level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VIII.II Alternative Measures of Natural Disaster Intensity

In the baseline analysis, I use the average wind speed to capture natural disaster intensity at the MSA-month level. As a robustness check, I instead define disaster intensity using the maximum

monthly wind intensity observed within each MSA, although this measure may be more sensitive to the timing of a single extreme wind event within a month. Table 18 shows that the main results remain qualitatively similar under this alternative definition. For Category 4+ hurricanes defined using the maximum wind speed measure, both the average treatment effect and the estimated interaction coefficients closely resemble the baseline estimates, indicating that the findings are robust to alternative measures of disaster intensity. For Category 3+ hurricanes, the average treatment effect is not statistically significant, although the interaction effects with industrial concentration, road connectivity, and intra-city goods sourcing remain statistically significant and retain the same signs as in the baseline specification.

Overall, these results suggest that the baseline average-intensity measure provides a more conservative definition of disaster exposure by requiring severe winds to be more broadly experienced within an MSA. For the average wind intensity of an MSA to exceed a given hurricane-category threshold, severe winds must affect a sufficiently large portion of the MSA. In contrast, the maximum-intensity measure classifies an MSA based on the strongest wind observed at any single location on any day within a month, even if such extreme winds are highly localized. Consistent with this distinction, the baseline Category 2+ results most closely resemble those obtained under the maximum-intensity Category 4+ definition. This similarity suggests that the two definitions identify disaster exposures of broadly comparable severity, despite differing substantially in how the spatial extent of exposure enters their construction.

Table 18: Robustness to an Alternative Disaster Intensity Measure: Maximum Intensity

		Change in Monthly MSA Electricity Load	
		Category 3+	Category 4+
<i>Panel A: Average Treatment Effect</i>			
Average treatment effect		−0.00598 (0.0103)	−0.0344*** (0.0030)
<i>Panel B: Interaction Coefficients</i>			
Labor Market	Industrial Composition (<i>HHI</i>)	−1.366*** (0.517)	−1.814*** (0.194)
	Road Connectivity (<i>wMEBC</i>)	3.634** (1.528)	3.540*** (0.748)
Trade Flow	Intra-city Goods Sourcing (<i>IntraTFR</i>)	0.0633* (0.0363)	0.130*** (0.0109)
Demographics	Population Density	−0.0000765 (0.000115)	−0.000185*** (0.0000546)
Observations		26,000	26,056

Notes: Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of the urban-structure characteristics and other covariates. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. Natural disaster exposure is measured using the maximum monthly wind intensity within each MSA rather than the baseline average monthly wind intensity measure. Columns correspond to Category 3+ and Category 4+ hurricane intensity thresholds. Standard errors, clustered at the MSA level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VIII.III Alternative LP-DiD Specification

VIII.III.1 Controlling for Pre-Period Exposure

In the baseline LP-DiD specification, I implement the LP-DiD specification and show that treated and control MSAs do not systematically differ in their exposure to natural disasters prior to treatment. More specifically, Table 4 shows no statistically significant differences between the two groups in either average or maximum pre-treatment wind intensity.

As a robustness check, I augment the baseline specification with controls for prior disaster exposure, measured using either average or maximum pre-treatment wind intensity. Table 19 shows that the estimated treatment effects and interaction coefficients remain highly consistent with the baseline findings. These results indicate that both the average treatment effect and the estimated heterogeneity across urban structures are robust to explicitly controlling for differences in pre-period natural disaster exposure across MSAs.

Table 19: Robustness to Pre-period Disaster Intensity Controls

		Change in Monthly MSA Electricity Load	
		Additional Control Variable	
		Avg. Pre-period Intensity	Max. Pre-period Intensity
<i>Panel A: Average Treatment Effect</i>			
Average treatment effect		−0.0314*** (0.0062)	−0.0311*** (0.0061)
<i>Panel B: Interaction Coefficients</i>			
Labor Market	Industrial Composition (<i>HHI</i>)	−1.608*** (0.262)	−1.602*** (0.259)
	Road Connectivity (<i>wMEBC</i>)	2.929*** (0.979)	2.962*** (0.956)
Trade Flow	Intra-city Goods Sourcing (<i>IntraTFR</i>)	0.0800** (0.0313)	0.0799*** (0.0307)
Demographics	Population Density	−0.000363*** (0.0000741)	−0.000361*** (0.0000726)
Observations		25,948	25,948

Notes: Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics and other covariates. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. The first column controls for prior disaster exposure using average pre-period wind intensity, while the second column uses maximum pre-period wind intensity. Positive interaction coefficients indicate that higher values of the corresponding characteristic attenuate the average decline in electricity load following natural disaster exposure, whereas negative coefficients indicate that they amplify it. Standard errors, clustered at the MSA level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VIII.III.2 Longer Stabilization Period

In the baseline LP-DiD specification, I also impose a one-month stabilization period prior to treatment onset. This restriction reflects the assumption that the effects of natural disasters are relatively short-lived, such that requiring cities to remain untreated for two consecutive months before a new disaster occurs is sufficient to ensure that they have returned to a clean untreated state. Nevertheless, this assumption may be restrictive if the effects of prior disaster exposure persist for longer, such that an MSA requires a longer untreated period before it can reasonably be considered to have returned to a clean untreated state. To examine the sensitivity of the results to this assumption, I extend the stabilization period by requiring no changes in status during the preceding L periods before treatment entry, thereby imposing a more conservative definition of a clean untreated state.

Table 20 examines the robustness of the baseline specification by extending the stabilization period from one month to between two and six months. Longer stabilization periods impose progressively stricter pre-treatment clean-window requirements by excluding observations that experience additional natural disaster exposure over increasingly longer pre-treatment intervals. Across

all specifications, the estimated treatment and interaction effects remain highly stable in both magnitude and statistical significance, while the number of observations declines only modestly as the stabilization period increases.

Table 20: Sensitivity to the Stabilization Length

		Stabilization Length (s)				
		$s = 2$	$s = 3$	$s = 4$	$s = 5$	$s = 6$
<i>Panel A: Average Treatment Effect</i>						
	Average treatment effect	−0.0307*** (0.00608)	−0.0308*** (0.00609)	−0.0308*** (0.00609)	−0.0307*** (0.00608)	−0.0308*** (0.00607)
<i>Panel B: Interaction Coefficients</i>						
Labor Market	Industrial Composition (<i>HHI</i>)	−1.559*** (0.273)	−1.556*** (0.273)	−1.555*** (0.273)	−1.553*** (0.273)	−1.553*** (0.273)
	Road Connectivity (<i>wMEBC</i>)	2.953*** (0.943)	2.951*** (0.943)	2.953*** (0.945)	2.951*** (0.943)	2.955*** (0.944)
Trade Flow	Intra-city Goods Sourcing (<i>IntraTFR</i>)	0.0682** (0.0319)	0.0681** (0.0320)	0.0681** (0.0320)	0.0680** (0.0319)	0.0681** (0.0319)
Demographics	Population Density	−0.000325*** (0.000075)	−0.000324*** (0.000076)	−0.000324*** (0.000076)	−0.000325*** (0.000076)	−0.000325*** (0.000076)
Observations		25,989	25,960	25,931	25,902	25,873

Notes: Panel A reports the marginal treatment effect of natural disaster exposure in the event month ($h = 0$), evaluated at the sample means of all urban-structure characteristics and other covariates. Panel B reports the estimated interaction coefficients between natural disaster exposure and each urban-structure characteristic. Each column varies the stabilization length s , defined as the number of consecutive pre-event months during which control MSAs are required to exhibit no treatment-status transitions. Standard errors, clustered at the MSA level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

IX Further Implications: Industry-to-Industry Mobility and the Propagation of Disaster Impacts

The industry-specific results in Section V reveal substantial heterogeneity in how concentration across industry groups shapes the economic consequences of natural disasters. These findings characterize how individual components of an MSA’s industrial composition contribute to its vulnerability to disaster shocks. The implications of these industry-specific vulnerabilities may extend further, however, once cross-industry connections are taken into account. If establishments in a particular industry are routinely connected through movements originating from a broad range of other industries, disruptions associated with that industry’s vulnerability may have consequences beyond the economic activity occurring directly within the industry itself.

To explore this broader dimension, I examine industry-to-industry mobility patterns based on individuals’ routine movements between establishments. The purpose of this exercise is to characterize

how the six industry groups considered in the preceding analysis are embedded in everyday urban activity and, in turn, to provide additional context for interpreting the heterogeneous industry-specific resilience results. In particular, the analysis allows me to assess whether industry groups associated with greater disaster vulnerability also occupy prominent positions in routine movements across otherwise different parts of the local economy.

IX.I Measuring Industry-to-Industry Mobility

I use device-level foot-traffic data from Veraset (2022) to construct industry-to-industry mobility networks. The data cover January 1, 2021 through September 29, 2022 and record sequences of establishment visits by individual mobile devices within an MSA over the course of a day. I order establishment visits within each device-day and identify transitions between consecutive stops. Each establishment is then assigned to one of the six industry groups used throughout the analysis. For example, a device observed moving from home to an RUC establishment, then to a PS establishment, and subsequently returning home generates three consecutive transitions, HOME-to-RUC, RUC-to-PS, and PS-to-HOME. To prevent device-days with longer observed itineraries from mechanically receiving greater weight, I normalize transitions within each device-day so that the weights assigned to all observed transitions sum to one. I then aggregate these normalized transition weights within each MSA-month.

For the analysis, I focus on mobility between economic establishments and exclude transitions involving home. For each MSA-month, I construct the conditional transition probability

$$P_{ij,mt} = \frac{W_{ij,mt}}{\sum_{k \in \mathcal{I}} W_{ik,mt}}, \quad i, j \in \mathcal{I}, \quad (12)$$

where $W_{ij,mt}$ denotes the aggregate normalized transition weight from origin industry i to destination industry j in MSA m and month t , and $\mathcal{I}$ denotes the six industry groups. Thus, $P_{ij,mt}$ measures the probability that the next observed economic establishment belongs to industry group j , conditional on the current establishment belonging to industry group i . For example, $P_{\text{HE,ATT},mt} = 0.3$ indicates that 30 percent of the normalized transitions originating from HE and ending at another economic establishment are directed toward ATT establishments. Finally, I average these transition probabilities first across months within each MSA and then across MSAs. This procedure gives each month equal weight within an MSA and each MSA equal weight in the aggregate network, preventing cities

or months with greater observed mobility activity from mechanically receiving greater weight. The resulting 6×6 transition matrix therefore characterizes the industry-to-industry mobility pattern of the average MSA in the sample.

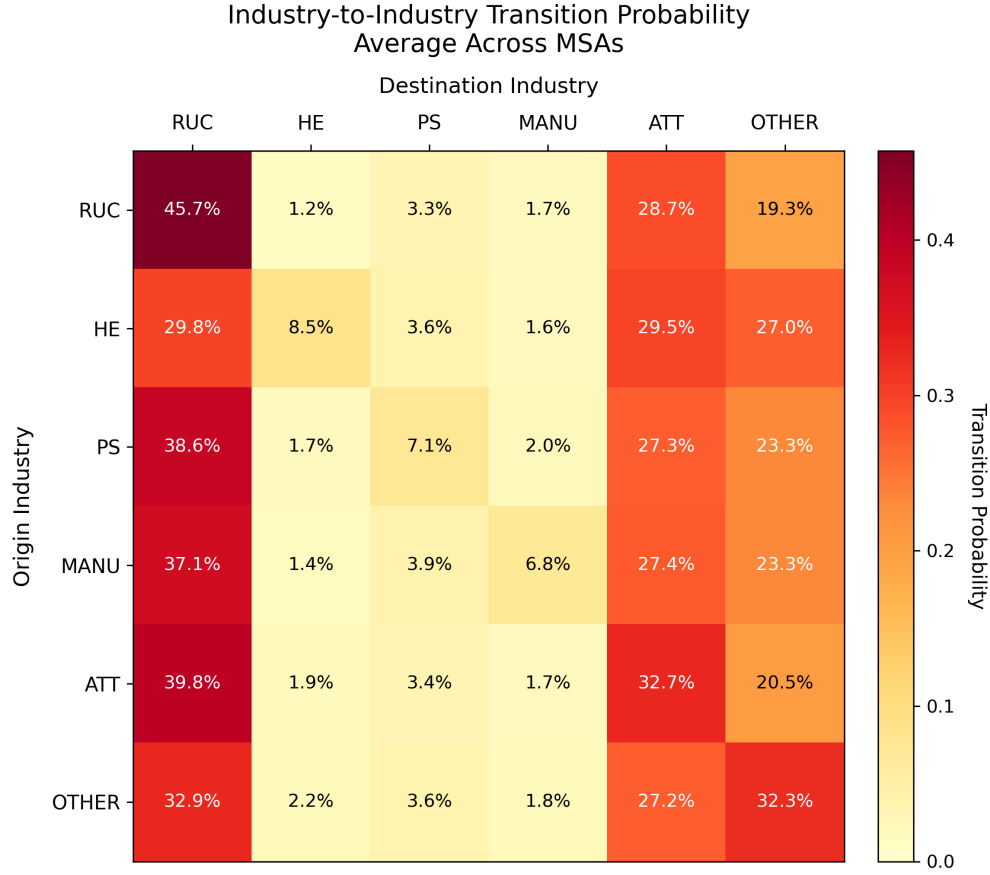
IX.II Mobility Patterns and Implications for Urban Resilience

The resulting mobility network reveals substantial heterogeneity in how individual industry groups are embedded in routine movements across the city. As shown in Figure 16, RUC and ATT stand out as the two most prominent destinations for movements originating from establishments across a broad range of industries. Regardless of the industry group from which a trip originates, a substantial share of subsequent establishment visits is directed toward RUC or ATT establishments. In other words, individuals previously observed at HE, PS, Manufacturing, or Other establishments are frequently observed moving next to RUC or ATT establishments. This makes RUC and ATT prominent components of the routine mobility network of a city.

For RUC, this pattern is intuitive because the group includes retail and other activities that are routinely used by households and workers throughout the local economy. Grocery shopping and other retail activities, for example, are relevant to individuals engaged in many different industries. ATT also plays a prominent role, but for a somewhat different reason. Accommodation, transportation, wholesale trade, and related activities are closely connected to movements originating from many different parts of the local economy. Thus, ATT is not simply important for individuals working in ATT industries; it is also a frequent destination for individuals arriving from many other types of establishments.

While Figure 16 summarizes the average industry-to-industry mobility pattern across all MSAs, an important related question is whether the prominence of RUC and ATT as mobility destinations persists when examining MSAs conditional on their industrial composition. More generally, the aggregate mobility pattern may mask systematic differences in industry-to-industry transitions across MSAs with different industrial compositions. I therefore examine how these mobility patterns vary with local industrial composition. For each of the six industry groups, I identify MSAs in the top quartile of the corresponding industry-specific HHI measure, HHI_{gc} , and separately average their MSA-level transition matrices. This procedure yields six comparable mobility networks for MSAs characterized by high concentration in RUC, HE, PS, Manufacturing, ATT, and Other Industries, respectively. Comparing these networks with the average mobility pattern reveals whether the prominence of particular industries in everyday mobility is specific to MSAs with greater concen-

Figure 16: Industry-to-Industry Transition Probabilities Across Cities

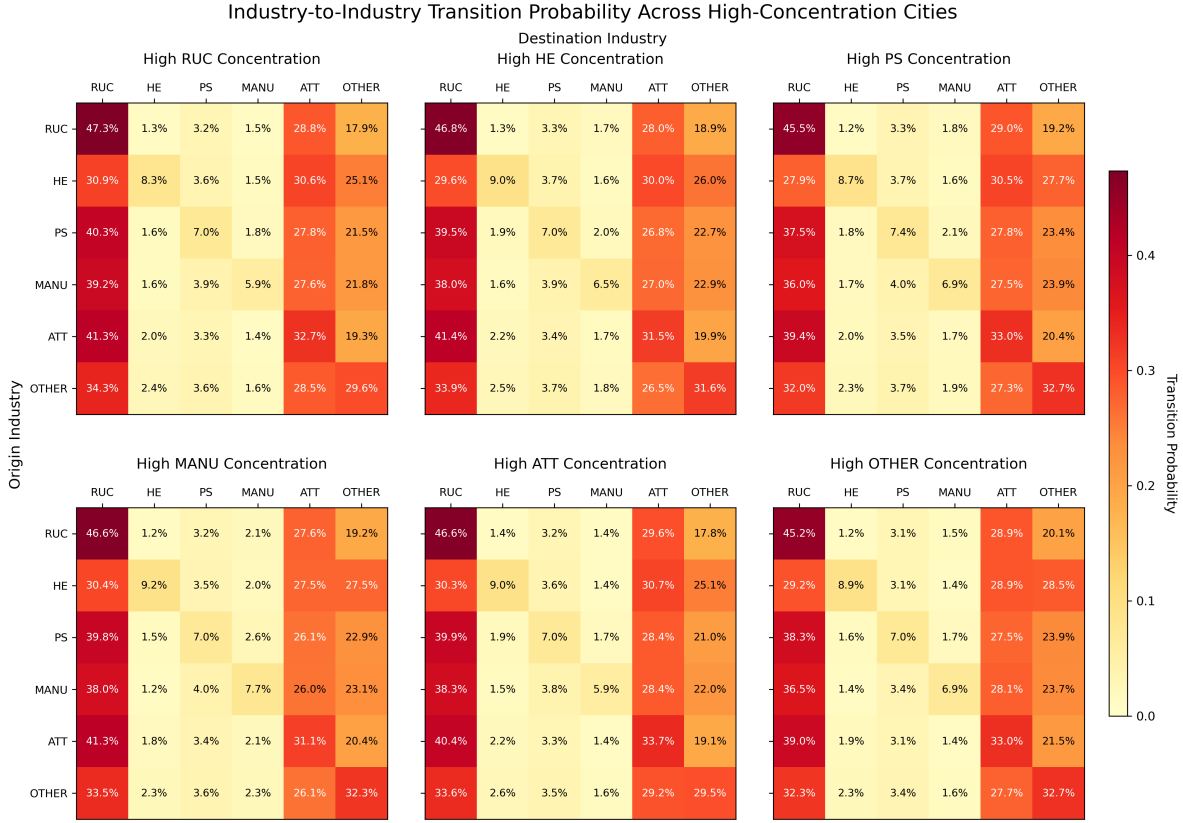


Note: This figure reports conditional transition probabilities across industry groups, excluding home-related transitions. Each cell shows the probability that the next observed non-home stop belongs to the destination industry indicated by the column, conditional on the current stop belonging to the origin industry indicated by the row. RUC and ATT emerge as the most prominent destinations across a wide range of origin industries.

tration in those industries or instead persists across different industrial compositions as a broader feature of urban economic activity. Figure 17 shows that the prominence of RUC and ATT is not specific to MSAs concentrated in these industries. Instead, both remain prominent mobility destinations across MSAs characterized by substantially different forms of industrial concentration. Cities with high concentration in Manufacturing, HE, PS, RUC, and ATT all exhibit substantial mobility toward RUC and ATT destinations. The prominence of these industries in routine mobility therefore appears to be a broad feature of urban activity rather than one specific to cities concentrated in those industries.

This mobility pattern provides a network perspective on the main finding that a greater ATT employment share exacerbates the economic consequences of natural disasters. Because ATT activities are closely connected to routine movements originating throughout the local economy, cities with

Figure 17: Industry-to-Industry Transition Probabilities Across Highly Specialized Cities



Note: This figure reports industry-to-industry transition probabilities separately for cities in the top quartile of concentration in each industry group. City concentration is defined using the industry group-specific *HHI* measure. The common transition structure is broadly preserved across concentration groups, with RUC and ATT remaining prominent destinations in each type of city.

greater employment concentration in ATT have a larger share of economic activity concentrated in a sector that is also deeply embedded in everyday urban mobility. Disruptions to ATT-intensive activity may therefore coincide with disruptions across interconnected parts of local economic activity, rather than remaining confined to the sector itself. The mobility patterns thus suggest that the vulnerability of ATT-intensive economies may have broader implications given ATT's prominent position in routine urban mobility. This mobility dimension is particularly informative when comparing the vulnerability of ATT- and HE-intensive economies. The main analysis finds that larger employment shares in either ATT or HE exacerbate the economic consequences of natural disasters, yet the two industry groups occupy notably different positions in the everyday mobility network. ATT is a prominent destination for movements originating from nearly every industry group, whereas HE does not exhibit the same degree of cross-industry connectivity. Thus, while greater sectoral scale in both ATT and HE is linked to greater disaster vulnerability, ATT is distin-

guished by its broader connections to routine activity across the local economy. The vulnerability of ATT-intensive economies may therefore have implications for urban economic activity beyond those captured by employment concentration alone. This distinction also informs the potential allocation of external resilience support. In ATT-intensive cities, public investments or post-disaster assistance that maintain or rapidly restore access to ATT-related activities may generate benefits beyond the directly affected sector, given their broad connections to routine economic activity throughout the city.

X Conclusion

Suppose a widespread winter storm, such as the January 2025 Blair event, simultaneously disrupts multiple metropolitan areas. Which cities experience larger economic losses solely because their underlying economic structures differ? Although urban agglomeration has long been recognized as a fundamental driver of economic growth, much less attention has been paid to how different forms of economic structures shape cities' vulnerability to natural disasters¹⁸. Given that most natural disasters are geographically localized, this remains an important gap in understanding the full economic consequences of such events. Because different concentration structures can either amplify or mitigate the magnitude and persistence of disaster-induced disruptions, their effects should be explicitly incorporated into comprehensive assessments of urban disaster impacts. Incorporating disaster resilience into evaluations of urban economic structure, therefore, provides a more complete perspective on the costs and benefits of agglomeration.

Given that cities should be assessed not only by the productivity gains generated through economic concentration but also by their capacity to withstand and recover from natural disasters, understanding these trade-offs is increasingly important for designing urban policies that promote both economic efficiency and long-term resilience. Urban development decisions have often prioritized the productivity gains associated with the concentration of economic activities while placing less emphasis on the disaster risks that such concentration may entail¹⁹. As natural disasters become more frequent and severe, identifying which forms of concentration amplify economic disruptions and which facilitate recovery can help local governments better balance the benefits of concentration

¹⁸For instance, Lin et al. (2024) show that, despite growing awareness of sea-level rise, people continued to move into coastal U.S. cities between 1990 and 2010 to capture the economic benefits of urban proximity. This illustrates the willingness to accept greater disaster exposure in exchange for economic opportunities.

¹⁹For example, Balboni (2025) investigates the long-run costs of infrastructure investments under climate change when future climate risks are not adequately incorporated into local investment decisions

against disaster risk when designing ex-ante policies, including infrastructure investment, land-use planning, and resilience strategies.

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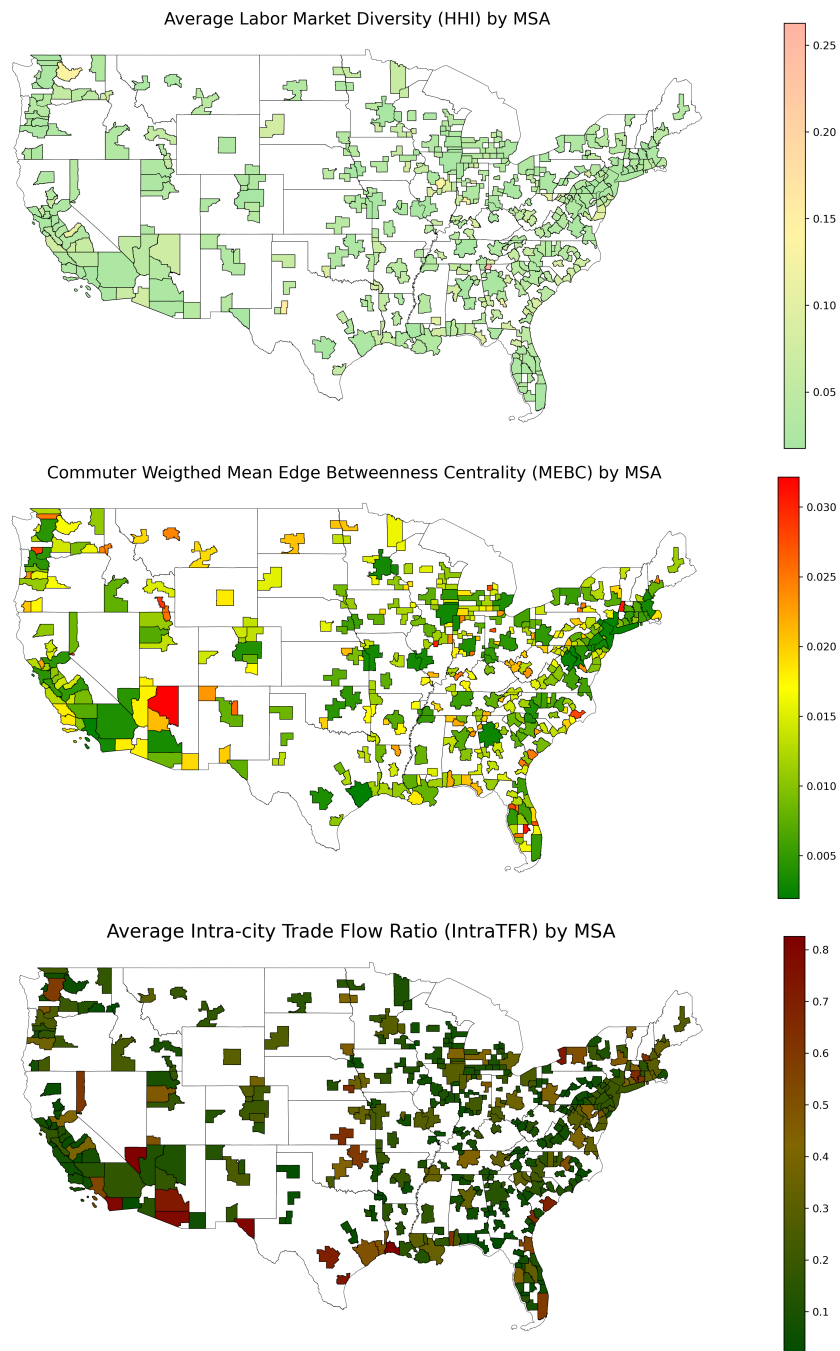
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XI Appendix

Figure 18: City-Level Concentration Measures (2017)



Note: This figure presents average concentration measures for the labor market, road network, and trade perspectives across MSAs from 2017.